# USE OF LIBRARY

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**Aims and Objectives**

The aims and objectives of this chapter is to acquaint students with what the library is, the types of libraries that we have, the components of a library and the various methods by which library resources are acquired into the library. This would enable students to be familiar with the library setup and be able to know which library is obtainable in any particular institution.

**At the end of this Chapter:**

- Students should be able to explain, in their own words, what the library is
- The types of libraries that we have
- The components (stock) of the library
- The various acquisition methods etc.

**What is a Library?**

The term library, derived from a Latin word 'liber' which means book, has been seen differently by some scholars. Some assume that it is a mere collection of books, housed in a room without any form of organization. Others felt that the library is just a reading room where people go for one information need or the other.

The Librarians' Glossary (1977) as cited by Busayo (2004, p.1) defined the library as follows:

- A collection of books and other library materials kept for reading, study and consultation
- A place, building, room or rooms, set apart for the keeping and use of a collection of books etc
- A collection of films, photographs and other visual non-book materials, plastic or metal tapes and discs, computers, tapes and programmes.

However, Jegede (2007) opined that the library in any establishment or institution is the store house of knowledge. Kumar (2011,p.3) quoted Ranganathan as defining a library as 'a public institution or establishment charged with the care of a collection of books, the duty of making them accessible to those who require the use of them and the task of converting every person in its neighbourhood into a habitual library goer and reader of books'. Thus, a library is regarded as a public institution, which is also expected to

convert the potential readers into actual readers.

In other words, it is the repository of information and information sources. The library can therefore be seen as an educational resource centre and of course, a 'nest' where scholars are hatched. The library is a store house of knowledge for students, lecturers, researchers and other scholars that may be allowed to use the library. The major roles/functions of the library are: collection of materials, organization, storage and dissemination of information resources.

**Why the study of Use of Library?**

1. The use of the library keep users acquainted with library resources and how to judiciously and maximally make use of these materials for academic benefits.
2. No would be successful scholar, would ignore or alienate the library, as the resources housed are needful and helpful to be a successful scholar.
3. The Use of the Library (GST 103) is a requisite course for all undergraduates. Students are therefore expected to register for this course, write the examination and pass it before they could graduate.
4. The study of use of library, no doubt, is a good take off point (foundation), for a would be successful academic, scholar or researcher.

## TYPES OF LIBRARIES

Busayo (2004) observed that there are six types of libraries in the society. He stressed that each type of library is unique and performs given roles. Some functions are common to all libraries, while others are tailored to meet the special requirements of the owners. The six (6) types of libraries are: Academic Library, National Library, Private Library, Public Library, Research or Special Library and School Library.

- **Academic Libraries**

Academic libraries are those that are attached to tertiary institutions of the status of a university, a polytechnic, a college of education or other tertiary institutions with similar standing, offering Certificate programmes, Diploma and Degree courses at various levels. Examples are FUOYE Library, EKSU Library, Federal Polytechnic Ado-Ekiti Library, College of Education Ikere-Ekiti Library etc.

Aina (2004) observed that the main purpose of a university library (which is an academic library) is to support the objectives of a university, which are in the areas of learning, teaching, research and service. The users of a university library are students, researchers, lecturers, support staff (non-teaching staff) and members of the local community in which the university is located. The library is regarded as the heart of the intellectual system of the university. To a large extent, the quality of a university is measured by the services provided by the library, because of its unique position in the over-all university system.

- **National Libraries**

National library is defined as the apex library of any nation. It is owned by a sovereign nation or state. It caters for the information needs of the country and it has legal deposit right, which is the right to acquire publications published on the country by individuals or organizations within or overseas, no matter their nationality, for retention in its stock for use of the citizenry. It is also empowered by law to issue: International Standard Book Number (ISBN) and International Standard Serials Number (ISSN), for books and journals respectively, to intending authors and publishers on request. Examples of national libraries are the National Library of Nigeria, National Library of Kenya and the Library of Congress of the United States.

- **Private Libraries**

These are libraries owned by an individual or an organization like a Church. They are established to meet the needs of the owner and thus the collections are dictated by the fields or disciplines of the owners. For example, some lawyers have their private (personal) law library; also some medical doctors have their private medical library. Likewise some engineers, librarians etc have private (personal) library to meet their reading needs. The stock usually is based on the profession of the owner and managed accordingly by the owner.

- **Public Libraries**

Like the name suggest, public libraries take care and serves the public interest of all ages without any discrimination. It is often defined as the public or peoples' university. It is usually financed with the tax-payers money and it makes reading provision for all in the community/society where it is situated, including the children. Most public libraries make provision for children reading section.

- **Research or Special Libraries**

Research or special libraries are designed for a defined or a unique group of users with similar needs. The users of research or special libraries are specialist-users, like the Law Libraries for Lawyers, Medical Libraries for Doctors and Engineering Libraries for Engineers etc. Organizations with such libraries are International Institute of Tropical Agriculture (IITA) Ibadan, Central Bank of Nigeria (CBN) Abuja, Nigeria Institute of International Affairs (NIIA) Lagos, Cocoa Research Institute of Nigeria (CRIN) Ibadan, Nigeria Institute of Social and Economic Research (NISER) Ibadan etc.

- **School Libraries**

School libraries are libraries attached to Nursery, Primary and Secondary Schools. It is intended to encourage reading habit and to develop in pupils, the ability to learn from books with or without the aid of a teacher. The importance of school libraries cannot be over-emphasized as it helps to develop reading skills and the love for books in children, early in their educational career. This is in line with the adage that says that

habits die hard and that Biblical phrase, 'Teach a child the way he should go, and when he is grown up, he will not depart'.

## ATTRIBUTES OF A LIBRARY

Some attributes/conditions/qualifications are required to be fulfilled or met before a place can indeed be seen or described as a library. Nwalo (2000) observed that many people in Nigeria abuse the use of the term 'library'. Often, one notices that rooms or personal offices in Nigerian schools or other public organizations are labeled 'libraries' simply because a few books or serials, no matter how old and irrespective of whether or not they are needed by anybody within the school or organization, are kept there. What a layman in our society calls library is no library but mere collection of books, often acquired with no specific policy and administered in a manner in contradiction of the purpose of the collection. Likewise, once a collection is administered by a person, who, no matter, his qualifications and position in society, has no formal education and training in librarianship, it cannot be regarded as a library. Such a person has neither the competence to acquire and organize library collections nor the competence to provide readers services.

Below are the known attributes of a library

1. There must be a convenient space or accommodation set-aside for no other purpose but library operations.
2. There must be a dynamic collection of books and other ephemeral materials in the library space.
3. The materials must be systematically organized in a useful order, stored and preserved for posterity.
4. Only a person who has undergone formal education and training in librarianship (Librarian, Library Officer or Teacher-Librarian) must be in charge of the library.
5. People must be allowed to use the library at specified time. In other words, there must be opening and closing hours and library users.

Keeping a collection, no matter the quality and quantity, out of the reach of prospective users negates the concept of library. Libraries have a common objective of acquiring information materials such as books, newspapers, journals, magazines, photographs, maps, films, microfilms, CD-ROMS and other forms of records. These are systematically organized, processed, preserved and disseminated to users in order to meet their educational, research, information and recreational needs.

## COMPONENTS OF THE LIBRARY

A typical library is made up of four (4) components namely:

- Books, Serials and Non-Book Materials
- Building or Room or Rooms
- Personnel (Staff)
- Clientele (Users)

## BOOKS, SERIALS AND NON-BOOK MATERIALS

- **Books** are printed materials of not less than 49pages, apart from the cover pages, that are fastened together for ease of reading, consultation or reference. When it is less than 49pages, it's known as a pamphlet.

- **Serials** are periodical publications which are issued in successive parts and they are intended to be continued indefinitely. Serials are either published at regular or an irregular interval, which could be daily, weekly, bi-monthly, monthly, quarterly or yearly. Serials usually bear volume, number and year ie 60 (2) 2016 eg daily newspapers, journals, magazines, annuals etc. It is more current than books and it is useful for research.
- **Non-book materials** are otherwise known as audio-visual materials. These are information resources that can be seen and or heard. They are not printed in readable form like books. Examples are:
- Computers
- Films
- Video-cameras
- Televisions
- Radios
- Slides
- Maps, Charts, Photographs etc

## BUILDING OR ROOM OR ROOMS

Building or Room is an important component of a library. There should be a designated building or room or rooms set aside as the library, where the books (collections) can be safely kept, where staff can function effectively and which users can conveniently make use of, without any form of noise or distractions, for maximum concentration and assimilation. The building or room must be conducive to reading, with good lighting, well ventilated, with air conditioners or fans etc

## PERSONNEL (STAFF)

Books, serials and non-book materials (collections) and building or room alone do not make a good/complete library. The personnel constitute an important component of a library. The staff mans the library and translates all the policies of the library into action, that is, they ensure the smooth administration of the library.

(1) The core library personnel consist of:

- Academic Librarians
- The Para-professional staff- The Library Officers.

(2) The supportive library personnel are:

- Library Attendants/Assistants, Porters, Clerical Officers etc.
- Technical Staff of different categories

**CLIENTELE (USERS)**

- Clientele, users, readers and patrons are terms that are used interchangeably to mean the same thing.
- The library is expected to provide information, to be disseminated by the personnel and used by the clientele.
- The library serves different users depending on the type of library and the need of the users. An academic library for example would serve students (undergraduate and post-graduate), lecturers, non-teaching staff and other researchers who may be allowed to use the library for reference services only.

These components are of equal importance. In other words, all of the four components work jointly and are inseparable from one another. The roles are complimentary (Symbiotic).

**STOCK OF THE LIBRARY**

The stock of a library refers to the various resources that the library house (keep) and makes available to the library users on demand. Thus, Library stock, like books, serials and non-book materials are acquired through different methods/channels namely:

- Purchase
- Gift or Donation
- Exchange
- Bequeath/Will

**Purchase**

This is otherwise referred to as 'cash and carry'. The librarian can decide to visit the book vendor directly and make his purchases or invite the book seller to his institution library for selection of needed titles to be made and then bought. Some book vendors also hawk, that is, they go from institution to institution, introducing their collections to libraries for direct purchases to be made.

**Gift or Donation**

Some philanthropist, authors, institutions, organizations etc donate books and other learning resources to library at will or when they are being solicited for. Such method/channel of acquisitions is referred to as gift or donation. These categories of materials come freely to the library without being bought.

**Exchange**

Exchange is another method/channel of acquisitions. That is, another way by which materials is being acquired into the library. It is not all gift or donated items that are of direct relevance to the beneficiary, hence an institution who received gifts or donations in a discipline that is outside her curriculum would politely accept such, and can decide to exchange such materials with other sister institution that offers the course and needs the materials. Usually, gift or donation is not normally rejected as there could be exchange with other institutions.

**Bequeath (Will)**

Bequeath (Will) is another method/channel of acquisitions. Some educationist or learned people in the society normally indicate in their will, that their entire library holdings should be donated to certain institution, may be, their alma mater or others, after their death. Such an arrangement is known as bequeath/will. For example, the Late Chief Obafemi Awolowo bequeathed his entire law collections to Ogun State University now Olabisi Onabanjo University in his will, before his death and this came into effect when he died in 1987.

**LIBRARY SKILLS AND LIBRARY RULES AND REGULATIONS**

Library users are expected to possess some basic skills in the use of the library and be accustomed with the ethics of the library. The library as an institution is governed by rules and regulations which are intended to guide the conduct and behaviour of users. Hence, users are expected to be familiar with the rules and regulations and abide by them. Usually, where there is no law, there is no offence. Therefore, users who violate any of these rules and regulations are liable to being sanctioned or penalized.

**Some of these rules, which equally enable the library to achieve its set goals and objectives, are:**

1. Every library user must be registered with the library to be able to make use of the library resources.
2. Registered library users are issued library identity card which grants them access to the library.
3. Silence must be maintained in and around the library premises.
4. Users must show their identity card to the library officials for identification, at the entry point while coming to the library.
5. Orderly conduct must be maintained at all times in all sections of the library.
6. The use of naked light is prohibited in the library.
7. Smoking, eating or drinking is not permitted in the library.
8. Marking or defacing of library books or any other library materials is not allowed.
9. All readers must show any books and papers they are carrying out, to the library officials at the control point (entrance) for checking.
10. Seats cannot be reserved by users for other users.
11. Pets cannot be brought into the library.
12. Indecent dressing is not allowed in the library
13. Materials leaving the library on loan must follow due procedure and be properly loaned out.
14. Legally borrowed materials must be returned on or before the date specified on the book card.
15. Returned (discharged) books must be delivered at the loan or circulation desk for proper documentation.

16. It is an offence to keep library books beyond the date specified for return as this would attract fine.
17. No materials in the reference, serials or reserved sections may be borrowed out of the library.
18. All consulted books must be left on the reading table. On no account should any library user shelve any book.
19. Any person who is suspected to be a security risk may be ordered out of the library.
20. Users are not normally allowed into the section of the library marked, 'For Staff Only', 'Private' or 'Reserved' except by permission
21. The use and charging of cell phones is not allowed in the library.

**Sanctions for violation of Library rules and regulations**

Users who violate any of the library rules and regulations are liable to being sanctioned or penalized. However, sanction or penalty would depend on the gravity of the offence committed. Users who committee grievous offences, would be summoned before the Library Disciplinary Committee for interrogation and thereafter, recommended for appropriate disciplinary action or sanctions. Such measures would normally serve as deterrent to other users.

Sanctions, fines or penalties could range from:

1. Writing of a letter of undertaking by users, to be well behaved
2. Replacement of book lost or payment in lieu.
3. Suspension from the use of the library for certain period of time
4. Dismissal from the university after being tried and found guilty

**Revision Questions**

1. In your own words, define a library
2. What is an academic library?
3. Distinguish between an academic and a school library
4. Define a book
5. What are serials?
6. Identify 5 components of a library
7. Name 3 methods by which resources can be acquired into the library
8. Explain the term Bequeath
9. Explain the importance of the circulation section of the library
10. Who is the Circulation librarian?

# SERIALS, REFERENCE SOURCES AND SERVICES, AND INTELLECTUAL PROPERTY RIGHTS

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## Introduction

Library achieves its goal of information provision for teaching, learning and research through acquisition, processing and dissemination of information materials to clients. These materials are in different categories of which two of them are serials and reference sources.

This chapter presents a detailed description of the information materials and how they could be distinguished from one another. It also captures the concept of intellectual property rights with emphasis on copyright; how copyright is violated and how not to infringe on author's rights when using intellectual products for assignments, projects and other literary works.

## Learning Objectives

At the end of this chapter, students should be able to:

- Define serials;
- Differentiate between serials and periodicals;
- Give examples of the two types of reference materials;
- Describe reference sources;
- Discuss Intellectual Property Right and
- Highlight and explain referencing styles

## TYPES OF LIBRARY INFORMATION MATERIALS

### 1. Serials

Serials are the vital information sources housed in the serial section of the library. Serials are published in discrete parts at regular or irregular intervals. They can be defined as publications issued in series with numerical or chronological designation and intended to be published indefinitely. Examples of serial collections are journals, conference proceedings, magazines, newspapers, newsletters, bulletins. Those serials that are published at regular stated intervals of less than a year (e.g. daily, monthly, quarterly, etc) are called Periodicals. Examples include journals, magazines, and newspapers. Serials could be in print form or it could be in both print and electronic forms. Serials are

published by institutions, learned societies, publishers and /or non- governmental organizations.

Serial publications usually have a title, volume, issue or number, date and International Standard Serials Number (ISSN). The information in serials are original, current and up to date. Serials are consulted only within the library, and cannot be borrowed by library users.

**a. Journals**

This periodical contains original research reports or recent developments in a particular discipline. First, articles are submitted to Editors, who in turn arrange for peer review by experts in the field of submitted manuscripts. After receiving the reports of reviewers, decisions are arrived at by the editors as to whether to accept or reject manuscript for publication. The reviewing process is to ensure that standard is maintained in what is being published. Journals are published at regular intervals such as quarterly, bimonthly, monthly, etc. (example: African Journal of Library, Archives and Information Science published twice in a year)

**b. Magazines**: A magazine is a periodical which contains articles on different topics. The articles are usually concise, informal and not necessarily written by experts. They are educative and informative; usually, they contain graphics to arouse the reading interest of the public. Magazines can appear in print or both print and electronic forms. Examples include Tell, Newswatch, Ovation among others.

**c. Conference Proceedings:** These serial publications contain the proceedings (records) of a conference. Most times, the full text of the papers presented at a conference is published even though not all conferences publish proceedings.

**d. Newspapers:** These are periodicals which contain news of current events, advertisements, people's opinions and other current issues. They are usually available in print and digital format. Examples include Vanguard, Tribune, etc.

**e. Newsletters:** These publications are smaller in size than newspapers. They contain news on a particular organization or activity. They may appear in print or electronic format.

**f. Diaries:** Diaries are usually handwritten; it is an account of events in a person's life. Scientists write down their observation or progress made on their research work from time to time. Therefore, other scientists who may want to repeat such experiments will find the diaries useful.

**2. Reference Sources**

Reference sources are sources which people refer to or consult for quick and specific information in the library. Gates (1994) as cited by Ojedokun (2007) described a reference source as a source which is consulted for aid or information on a topic, a theme, an occurrence, a person, a date, place or a word. Reference sources can be used to gain background knowledge on a topic. They provide direct information and are not read

from cover to cover. Some of them are access tools which guide or direct on how to locate other information sources. Reference sources may be in print and /or non print format. These sources are not loaned out but consulted in the library. Reference books in the library carry the inscription "R" or "Ref" on their spines.

**Types of Reference Sources**

There are two types of reference materials.

(1) Directional reference materials

(2) Non –Directional reference materials

**Directional reference materials**: These sources direct users to where information can be obtained. e.g. bibliographies, indexes, abstracts.

**Non-Directional reference materials**: In this case, the required information is obtained directly from the sources e.g. directory, yearbook, encyclopedia, dictionary, handbook, map, gazette, gazetteer, atlas, manual, almanac, biography.

**Directional Reference Materials**

**a. Bibliographies:** Bibliographies are compilations of published works (i.e. books and articles) by an author, works on a particular discipline, topic, works published in a particular country or within a particular time frame. The published works are listed alongside their bibliographic details and this facilitates retrieval of the needed information material. Bibliographic details include the author, year of publication of work, title of work, edition statement, title of journal (where applicable), place of publication, publisher, pagination (where applicable). Example of Bibliography: Bibliography of Science and Technology.

**b. Indexes and Abstracts:** Indexes and abstracts are finding tools for tracing articles, books or other information sources. Indexes contain keywords or index entries while abstracts contain summaries of works such as books, article, speech, report, and dissertation. The information in the published abstract helps the researcher to decide whether requesting for the full text will meet his information need or not. They contain certain bibliographic details (e. g. author, title) for easy retrieval of relevant works. Examples: Library and Information Science Abstract (LISA), Dissertation Abstract International, Social Sciences Index, Science Citation Index.

**Non-Directional Reference Materials**

**a. Dictionaries:** This is a reference source with alphabetical listing of words. The listed words can be general words of a language (General dictionary) or specific words of a particular subject or discipline (Subject dictionary). Dictionaries give meanings, spellings, pronunciations, and usage of words. Some dictionaries give the derivation (etymology), synonyms, antonyms, and illustrations, biographical and geographical names of words. Example of a General Dictionary: Advanced Learner's Dictionary. Examples of a Subject Dictionary: Dictionary of Medicine, Dictionary of Music.

**b. Encyclopaedias:** This reference material contains articles on all aspects of human knowledge. They provide essay-like summary on a variety of topics .They are usually

arranged alphabetically and may appear in single or multiple volumes. Encyclopaedias are written by experts in various disciplines. Multivolume encyclopaedias usually have their indexes at the end of the last volume. General encyclopaedias contain information on all areas of knowledge while Subject encyclopaedias focus on specific disciplines. Examples of General Encyclopaedia: Encyclopaedia Americana, Encyclopaedia Britannica. Examples of Subject Encyclopaedia: Encyclopaedia of Architecture.

**c. Directories:** This reference source is a compilation of names of people, institutions, companies or organizations with their addresses, phone numbers, affiliation, publications and other details in classified or alphabetical order. Examples: American Library Directory, Nigerian Telephone Directory.

**d. Thesauri:** Thesauri (singular, thesaurus) contain list of words that has been arranged alphabetically. Thesauri list words with their synonyms and antonyms and other related words. It is worthy to note that thesauri do not give meanings of words. An example is Roget's Thesaurus.

**e. Gazettes:** These are official publications of the government which carry current information on the activities of the government. The publications are issued regularly and they include the constitution of a country, government appointments, Acts and Bills, Regulations, Tenders. An example is the Nigerian Gazette.

**f. Gazetteers:** These reference publications contain geographic names of places and their locations. They also contain the geographic features (e. g rivers, cities, lakes, mountains) of the locations. Example includes Chamber's World Gazetteer.

**g. Yearbooks:** This is an annual publication which contains statistics, photographs and other facts about events of the previous year. It covers past events of a country, discipline, or institution. Examples of Yearbook: Commonwealth University Yearbook.

**h. Almanacs** This is a collection of dates, facts and statistics which may be current or both current and retrospective. Almanacs are rich sources of economic, political, geographic and demographic information. It is an annual publication and is usually presented in tabular form to enhance comparison. There are general almanacs e. g World Almanac and Book of Facts, and Subject almanacs e.g. Almanac of American Politics.

**i. Handbooks:** They contain factual information in a brief manner on a particular subject. They are usually handy (easy to reach) and easily accessed e. g. CRC Handbook of Chemistry and Physics.

**j. Maps:** These referencematerials may be available in print or electronic form. They are abstract representations of the surface of the earth or parts of the surface of the earth. They serve as guides for travelers in locating their destinations. There are different types of maps e.g. street map, road map, relief map, topographical map, language distribution map and orthophoto map.

**k. Atlases:** An atlas is a bound collection of maps. The listing of the maps is done alphabetically. An example of an atlas is the World Atlas.

**Reference Services**

Reference services in the library include all the activities carried out by the reference librarian in satisfying or meeting a patron's information need. Such activities include answering users' enquiries, assisting or guiding users in locating information sources or resources, searching for information for patrons, helping to assess the quality or appropriateness of an information source, referral services ( i.e. referring patrons to resources in other libraries), and developing the reference collection. Other reference services include User Education which encompasses: acquainting new users with library sources, resources, their locations, services offered, the layout of library building and library rules and regulations. Compilation of Bibliographies, Provision of Current Awareness (CAS) and Selective Dissemination of Information (SDI) Services. These services can be rendered via e-mail, electronic mailing list, telephone, on-line chatting, phoning. For an effective service delivery, a reference librarian must be equipped with the following:

1. Reference Books
2. Computer with internet connectivity, databases and other relevant e-resources
3. Printed lists of library holdings that are not in the catalogue e.g. newspapers.

**Intellectual Property Rights**

Intellectual Property (IP) deals with the legal protection of intellectual property products. Intellectual property products are people's ideas e.g. musical compositions, the story in a novel. Intellectual property products are "the creative ideas and expressions of the human mind or intellect" (Bothma et al., 2014). They are intangible in nature. Intellectual products have value, and needs to be protected legally, thus, the need for Intellectual Property Rights. Intellectual property rights are the rights granted to an author or owner of intellectual products (i.e. ideas in books, films, etc).These rights vary across different countries, some countries focus on the economic value of the products for authors while some focus on the moral rights of users. IP applies to trademarks, patents and copyright.

**Trademarks:** Trademarks are symbols or signs put on products to show their commercial origin.

**Patent:** A patent is a legal document which establishes the right of an inventor of a product to use and derive financial gain from his invention for a specified period of time.

**Royalties:** This is a sum of money that is paid to owners of intellectual or creative works each time the work is sold or performed.

**Copyright**: This is the right granted by law to owners of intellectual and/or creative works to monopolise the use of their works or control its use by others. The symbol © represents copyright. Copyright laws vary across different countries, in Nigeria; copyright covers the following category of works

a. Literary works
b. Musical works
c. Artistic works

d. Cinematograph films
e. Sound recordings

**Exceptions to Copyright**

The use of other people's work is permitted under some certain conditions. This permission can be by law or under what is called "Fair use" or "Fair dealing". Fair use is the legal opportunity granted an author to cite a copyrighted work or make use of such work without obtaining permission from the owner of the work while fair dealing allows one to reproduce limited portions of copyrighted works. Fair use can be applied when conducting a research, criticism, news reporting, teaching and commentary while fair dealing includes works used for review, research or private study. Fair use of a copyrighted work is not for commercial purposes; also, a limited portion of the work which will not hamper the marketability of the work should be used.

REPRONIG (2004) states that copyright is not infringed if a work is performed, copied, reproduced or adapted for:

1. Research or private study
2. Criticisms or review or the reporting of current events if an acknowledgement is made.
3. The inclusion in a film or broadcast of an article work and shown where it can be viewed by the public.
4. Reproduced and distributed in a place where it can be viewed by the public.
5. The performance or broadcast of a reasonable extract, if sufficient acknowledgement made.
6. Reproduced and/or performed for judicial proceedings.

**Violation of Copyright Law**

Copyright is violated with the following activities:

**Plagiarism**: Plagiarism involves copying from an intellectual creation of another person without acknowledging or giving credit to the owners of the work. This is also called Academic Dishonesty. It should be noted that , even when using one's own previous work in another work, one still has to acknowledge himself and not self-plagiarize or infringe upon a publisher's copyright. Plagiarism is a grievous offence in academia and perpetrators of such acts are sanctioned accordingly.

**Piracy**: Piracy involves making copies of copyrighted works for financial gain. Such act deprives original owners of the works of his financial benefits and discourages him from producing more works.

**Counterfeits**: Counterfeits is the reproduction of fake copies of an original work for sale.

**Photocopy:** Photocopying a copyrighted work, if at all will be done for the purpose of research or private study, must be done reasonably, that is, only a copy of the work should be photocopied and a limited portion which will not hamper the economic gain of the author should be photocopied. Otherwise, one will be infringing or violating the copyright law.

**Referencing**

Referencing is "giving relevant, accurate and complete information indicating and acknowledging which sources you used for writing an assignment, a thesis or dissertation, a research article or a research report for work" (Bothma et.al., 2014).

Referencing, also known as "citing" or "citation", is usually done in a standardized manner according to the particular reference style you have adopted. Referencing shows that you have researched extensively on the topic you are writing, it indicates that your writing is not based on your ideas or view alone but other authors' views have been looked into, referencing also makes the tracing of the original works used by you possible for other authors.

When using someone else's work, you can quote the person directly (verbatim), paraphrase i.e. saying the same thing that another author has said using entirely different words. This involves digesting the whole idea and expressing it in your own words, or reflecting his original idea in your work. Anytime this is done, the owner(s) of the work must be cited in the text/document and also, a reference list must be presented at the end of the document. This reference list saves the author from being charged for plagiarism. When quoting verbatim, short quotation is always in quotation marks while long quotation is indented.

**Reference list:** A reference list is a list written at the end of your document that contains a comprehensive description of all the sources that you used, referred to or cited in the text of your document. Usually, a Reference list/Bibliography is alphabetical in arrangement, even though, in some cases, the arrangement can be a numeric one.

**Bibliography:** Bibliography is a word with different meanings. In this context, bibliography is synonymous to a reference list. It is a list of sources used and referred to in the document written by you. A bibliography can also be a list of information sources that you consulted but did not cite in your document.

**The Basic Principles of Referencing**

**In-text referencing**

In-text referencing involves mentioning (acknowledging) the source of the information you have cited or used in the text of your document. The details acknowledged in the text include the surname of the author cited and the year of publication of the work used. E.g. According to Okwilagwe (2001), copyright is the exclusive right given to an author and

other creative artists to authorize the use of their work in any way.

OR

Copyright is the exclusive right given to an author and other creative artists to authorize the use of their work in any way (Okwilagwe, 2001)

**Footnotes and Endnotes**

Footnotes are short notes that appear at the end of a page. It usually contains a small number or alphabet in the text that directs you to the footnote which either tells you the source of the information or gives more information about the text in the document.

Endnotes are usually found at the end of the document and they contain information about the text. However, it is worthy to note that, not all reference styles make use of footnotes and endnotes.

**Referencing / Citation Styles**

Referencing or citation is done using different citation styles. However, the type of style used depends on the specification of the publishing house /organization /faculty for which the work is written or the discipline of the writer. The different referencing styles are:

Harvard Citation Style: This style and its many variations are used in social and behavioural sciences, philosophy, politics, economics, arts and the health sciences.

APA (American Psychological Association) Style: This reference style is commonly used in the social sciences, education, and history.

MLA (Modern Language Association of America): This style is used by people in literature, arts, humanities and some other disciplines.

Chicago Manual of Style: This style is sometimes used by people in the social sciences, history, political studies, theology and computer science.

Turabian Citation Style: This style is a simplified version of Chicago Manual of Style. It was developed by Kate Turabian for the writing of theses, dissertations and term papers.

**Exercises**

1. What is a Serial Publication?
2. Give five (5) examples of Serials
3. What is the difference between Serials and Periodicals?
4. What are reference sources?
5. Give two (2) examples of Directional Reference materials and five (5) examples of Non-Directional Reference materials.
6. Define Intellectual Property Rights
7. Explain the following terms :
   (a) Trademark
   (b) Patent
   (c) Royalties
   (d) Copyright
8. Mention three ways through which copyright is violated

# BOOK ANATOMY (BIBLIOGRAPHIC PARTS OF A BOOK) AND BIBLIOGRAPHIC CITATION AND REFERENCING (GST103)

By

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## 1. Introduction (Course Description)

This chapter provides information on the bibliographic parts of a book. It also gives reasons for references and bibliographies in academic writing. It further provides insight into bibliographic citation and referencing, with specific attention given to the Harvard style of referencing, the APA style of referencing and the MLA style of referencing.

**Aim of the course**

At the end, students should be able to have thorough knowledge of different parts of a book and be able to appropriately cite references for any assignment/project carried out with APA, Harvard or MLA style of referencing. Specifically, you should be able to:

- Identify different parts of a book
- Know the difference between references and bibliographies
- Have knowledge of different referencing styles (e.g. APA, Harvard and MLA)
- Do in-text citation
- Correctly cite references and bibliographies

## 2. Book Anatomy (Bibliographic Parts of a Book)

A book can be defined as a medium for recording information in the form of writing or images, typically composed of many pages (not less than 49), bound together and protected by a cover. The technical term for the physical arrangement of book is called *codex*. A single sheet in codex is a leaf, while each side of a leaf is page. However, with the advent of information and communication technology (ICT), there is electronic version of book (e-book). E-book is a text-filled or illustrated pages produced in electronic format.

Bibliographically, the book can be divided into several parts. It is important to describe each part because the librarian refers to the various parts of a book when providing bibliographic description of each book. A book consists of two major parts. These are the preliminary part and the body of the texts. Each part has an important role to play. Parts of a book are as follows:

- **The Cover:** A book has two covers. i.e., the first cover which can either be 'hard bound' or 'soft bound'. A soft bound book is usually called 'paper back'. Hard bound books are more expensive than soft bound or paper-back books because of their durability.
- **The Spine:** this refers to pages of a book that sewed or glued together along one side. The title of the book and name of the author are printed on the spine.
- **The Dust Jacket:** some books have Dust Jacket. This is provided for a book to protect the cover of a hard bound book.
- **The End Paper:** This is the blank sheet that comes before the preliminary pages.
- **The Preliminary Pages:** These are the pages inside the front cover of a particular book; this part precedes the main text. The printed sheets in this section of a book are either unnumbered or numbered in roman figure rather than Arabic numeral which are associated with text.
- **The Half Title:** It contains information not as full as the title page.
- **The Title Page:** This is the page where the author's name, title of the book, publisher, place of publication and date of publication are found.
- **Verso:** This is the back title of the page. It gives the dates of the copyright, the name of the copyright owners, and other information.
- **Rector:** The right-hand page of a book usually bearing an odd page number.
- **Copyright (c):** This is referred to statement that the book's contents are property of the author or publisher.
- **Author:** A person who writes book.
- **Publisher:** This is the company which prepares and makes the book available to the public.
- **Year of Publication:** The year the book was published.
- **Place of Publication:** The country or town where the book was published.
- **ISBN:** The International Standard Bibliographic (Book) Number is a thirteen unique number given to a particular book by publisher so as to distinguish that book from any other book. It is a number of thirteen (13) digits put in certain pattern to identify a book, e.g. 978-81-313-0904-9. Every ISBN is given only to a single book title in the whole world.
- **The Preface:** This is where the author discusses briefly all aspects of the book.
- **Table of Content**: A table showing the order of information in the book.
- **The Text:** This is the main part of the book and it is divided into chapters.
- **Dedications:** Author's way of showing gratitude and appreciation to someone.
- **Acknowledgement:** Appreciation for aid or assistance rendered by others to author during the preparation of the book.

### a. Harvard System

The Harvard System, also sometimes called the author-date method because the main means of indicating a reference to someone else's work in the text is to insert the author's surname and date in the text and placed after Appendices if any. The citations should be divided into a list of References to works actually cited in the text and a Bibliography which lists the literature generally used in preparation for essay or project. It will not be necessary to list the references again in the bibliography. The lists should be arranged alphabetically by first author regardless of format. There is no need to separate out into books, journals etc.

References in the text are indicated by the name of the authors, (omit initial) and the year of publication in the parentheses e.g. Alaba (2007) if cited at the beginning of the sentence. When the author has published more than one cited work in the same year, these are differentiated by adding lower case letters after the year within the parentheses e.g. Alaba2007a, Alaba 2007b, Alaba 2007c etc. Repeated references are indicated in the text in the same way as the original ones.

### Harvard System Standard Layout for Citations

#### 1. Book

Surname initials (date). Title: Subtitle. Edition statement, Place of Publication, Publisher. For example:

**Olajide, O.**(2010). *A Perfect Guide to Basic Library Skills for Tertiary Institutions. Lagos*: Taisoft Publishers.

#### 2. Journal

Surname, initials (date). Article title. Journal title, volume (part), pages. For example:

Olajide, O. &Ariwodola, M. B. (2009). Utilisation of school libraries by teachers: a case study of public secondary school teachers in Ado-Local Government Area of Ekiti State, Nigeria. *Fountain of knowledge: Journal of Library and Information Science, 1*(1), 104-115.

#### 3. Two (or up to three) authors

Blair G M, Jones R S and Simpson R H (1975).Educational Psychology. 4th ed. New York, Wiley.

#### 4. Multi-authorship (4 or more authors)

Stone G M et al (2006) Principles of Biology. 2nd ed. New York, Macmillian Publishing Co. Inc.

#### 5. Books with editors or edited books

Akinrinade S. et al (ed.) (2004). Locating the local in the global voices on a globalized

Nigeria.Ile-Ife, ObafemiAwolowo University.

**6. No author or editor or anynomous works**

When this situation arises, use the name of the organization responsible for the work.
National Planning Commission (2004).National Economic Empowerment and Development Strategy. Lagos, NPC.

**7. Chapter in a Book**

If there is need for you to quote or cite a particular chapter of a book and it has a separate author, you can list it as follows: Adefolalu D. A. (2002). Climate. In Dominique Metaillet et al (ed.) (2002). Atlas of Nigeria. Paris, Les Editions J A

**8. Chronological Order**

When citing more than one reference at the same place is the text they should be listed in chronological date order i.e. earliest first.
Alokun (2000), John (2002) and Ajayi (2006).

**9. Government Publication**

Nigeria Federal Ministry of Education (1990). Blue print and action plan for the eradication of mass illiteracy by the year 2000. Lagos, Macmillian.

**10. Reference Books**

Familiar reference books can be referred to by their title unless there is an editor or complier to use instead e.g. Oxford Advanced Dictionary (1995). Oxford, Oxford University Press.

**11. Theses and Dissertations**

It is essential to include details of the level of degree etc. and awarding institution in the full details e.g.

Olajide, O. (2019). The role of school libraries in inquiry based approach for teaching and learning science subjects in senior secondary schools in Ekiti State, Nigeria. **(Doctoral thesis).**Department of Library and Information Science, **University of the Western Cape, South Africa. Retrieved from (Insert weblink)**

**12. Conference Paper and Proceedings**

For example, a conference paper and proceedings could be cited as:
Rachmawati&Ekowiyanti, M. I. (2016). ***The role of Indonesian school librarian in increasing students' literacy skills in digital age.*** **Paper presented at 45th Annual**

**Conference of IASL incorporating the 20th International Forum on Research in School Librarianship** held between 22-26 August at Meiji University, Tokyo, Japan.

**13. Newspaper Article**

Articles from Newspapers or magazines can be cited in the same format as for journals though there is need to include the exact date of publication e.g.:

Nejo, A. (2019, March 25). Fayemi names new schools after Fasanmi, Fasuan, others. *The Punch Newspaper*. Retrieved from https://punchng.com/fayemi-names-new-schools-after-fasanmi-fasuan-others/

**14. Information from the World Wide Web (Internet)**

Information derived from sites on the World Wide Web (www) can be treated in much the same way as printed sources. For example:

Raosoft Inc. (2004). Sample size calculator. Retrieved from: http://www.raosoft.com/samplesize.html

**b. APA Citation Style**

This refers to the rules and conventions established by the American Psychological Association (APA) for documenting sources used in a research paper. APA style requires two elements for citing outside sources:

**Reference Citations in Text and a Reference List**

Collectively, these elements identify and credit the sources consulted in the paper and allow others to access or retrieve this material. In APA style, citations to sources are placed in the text of the paper or project in order to briefly identify sources for readers and enable them locate the source of the cited information in the Reference list. The **author's last name** and the **year of publication** enclosed in parentheses as in case of Harvard System. For example: Olajide (2020) asserts or in a recent study of management (Olajide, 2020). References cited in the text of a research paper/project must appear in a Reference list or bibliography. This list provides the information necessary to identify and retrieve each source.

**APA Citation Style Layout**

**1. Book**

References to an entire book must include the following elements: Author(s) or editor(s), date of publication, title, place of publication and the name of the publisher. For example: **Olajide, O.**(2010). *A Perfect Guide to Basic Library Skills for Tertiary Institutions. Lagos*: Taisoft Publishers, 90pp.

**2. Journal**

Author(s) editor(s), date of publication, article title, journal title, volume number, issue number (if applicable), and page number. For example

Ahmadu, B.O. and Irochukwu, P.U. (2003) Development of an automated fund management system for Nigerian Universitites, Nigerian Libraries 37 (2), 1-16.

**3. Two (or up to three) author**

Blair, G.M., Jones, R.S., & Simpson, R.H. (1975).Educational psychology. 4th ed. New York: Wiley.

**4. Multi-authorship (4 or more authors)**

Stone, G.M. et al (2006). Principles of Biology. 2nd ed. New York: Macmillian Publishing co. Inc.

**5. Books with editors or anonymous works**

Akinrinade, S. et al (eds.). (2004). Locating the local in the global voices on a globalized Nigeria. Ile-Ife: Obafemi Awolowo University.

**6. No author or editor or anonymous works**

National Planning Commission (2004).National economic empowerment and development strategy. Lagos: NPC.

**7. Chapter in a Book**

Adefolalu, A.D. (2002). Climate .In M. Dominique et al. (Eds.) Atlas of Nigeria (pp. 62-65). Paris: les Editions J.A.

**8. Chronological order**

This should be cited and listed as in Harvard system.

**9. Government Publication**

Nigeria. Federal Ministry of Education.(1990). Blue print and action plan for the eradication of mass illiteracy by the year 2000. Lagos: Macmillian.

**10. Reference Books**

References for Dictionaries or Encyclopedia must include author(s) or editor(s) if known but if no by title, date of publication, title, place of publication, and the name of the publisher e.g. Oxford Advanced Dictionary. (1995). Oxford: University Press.

**11. Technical, Research Reports and Theses**

References to a report/thesis must include author(s), date of publication, title, place of publication, and name of publisher. In case of these or dissertations it is important to include details of the level of degree and awarding institution in the full details.

**12. Electronic Media and Online Sources**

APA's recommendations for citing electronic media call for consistent observation at least two important guidelines:

(i) direct readers as closely as possible to the information being cited, whenever possible, reference specific documents rather than home or menu pages;

(ii) provide addresses that work.

A reference of an internet source should provide title description, a "date" (either the date of publication, update or date of retrieval) and an address (in internet terms, a URL). For example:

Raosoft Inc. (2004). Sample size calculator. Retrieved from http://www.raosoft.com/samplesize.html

**13. Conference Paper and Proceedings**

For example, a conference paper and proceedings could be cited as:

Rachmawati & Ekowiyanti, M. I. (2016). ***The role of Indonesian school librarian in increasing students' literacy skills in digital age.*** **Paper presented at 45th Annual Conference of IASL incorporating the 20th International Forum on Research in School Librarianship** held between 22-26 August at Meiji University, Tokyo, Japan.

**c. MLA Citation Style**

The MLA (Modern Language Association) System of documenting sources which is set forth in the MLA Handbook for Writers of Research Papers, 6th ed. (New York: MLA, 2003). In MLA style, sources are acknowledged by short parenthetical citations in the text. This includes the author's last name and the page number cited. e.g. (Olajide, 85).

**MLA Citation Style Layout**

**Book**

**Olajide, Olabode**. "A Perfect Guide to Basic Library Skills for Tertiary Institutions". *Lagos*: Taisoft Publishers, 2010.

**Journal Article**

Kalu-Uche, N., Alamina, J. I. &Ovute, A. O. Pedagogical practices in the teaching of science in secondary schools in Rivers State, Nigeria.*Journal of Research in Humanities and Social Science, 3*, 2 (2015): 50-55.

**Newspaper or Magazine Article**

Nejo, Abiodun. "Fayemi names new schools after Fasanmi, Fasuan, others". *The Punch Newspaper*.(2019, March 25) Retrieved from https://punchng.com/fayemi-names-new-schools-after-fasanmi-fasuan-others/

**Chapter in a Book**

Adefolalu, Dayo A. "Climate" Atlas of Nigeria. Ed. Dominique Mataillet et al. Paris" Les Editions J.A., 2002.62-65.

**Encyclopaedia Article**

Sturgeon, Theodore. "Science Fiction".The Encyclopaedia Americana.International ed. 1995.

**Website**

Marcy, Bi. "Think They "Find Any Evidence of MallaryIruine?" Online posting. 30 May 2002: Mt. Everest> 99 Forum 28 June 2002 *http:/everstmountainzone.com/99/forum*

## 4. Conclusion

The citation styles discussed above are by no means better than the other. Citation formats that could be adopted depend on the discipline, institution, faculty and department. Whatever the method, strictly follow the laid down citation format adopted. In case of any doubt, consult with your lecturers, supervisors or editors to determine their referencing style preferred.

1. **Exercises** what is a book?
2. Pick any textbook of your choice and list all the bibliographic parts you can identify.
3. Differentiate between reference and bibliography?
4. Mention five (5) reasons for reference/bibliography.

## References

American Psychological Association (APA) (2020). APA 7th referencing: Getting started in APA 7th. Retrieved from https://libraryguide.vu.edu.au/apa-referencing/7GettingStarted

Barber, Graeme (2004) Har vard Guide to Citing References Retrieved July 10, 2007 from the World Wide Web: *http://www.marjon.ac.uk/library/books_journals/harvard-referencing.asp*

Hiers, J. T., Williams, J. O. and Arial, J. F. (1986). The Research Paper: Sources and Resources. Lexington: D.C. Hearth and Company.

**Olajide, O.** (2010). *A Perfect Guide to Basic Library Skills for Tertiary Institutions. Lagos*: Taisoft Publishers, 90pp.

(2004, Oct 29). MLA World Wide Web:

# USE OF LIBRARY

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**Course Description:**

- **ORGANIZATION OF LIBRARY**
- **LIBRARY CATALOGUE AND CATALOGUING**
- **CLASSIFICATION**
- **HOW BOOKS GET TO THE SHELF IN THE LIBRARY**
- **SECTIONS IN THE LIBRARY**

**Aims and objectives of the course**

At the end, students should be able to know the

- Various retrieval tools we have
- Different approaches to, importance and types of library catalogues
- Types of Classification Scheme we have and each alphabet for each faculty
- The various sections under each department

## Organisation of the Library

In any human setting, it is usual to organise objects that are within one's environment, especially those that are frequently used. This is to facilitate easy retrieval whenever the objects are needed. This is the principle of organisation. If there were no prior organisation of objects, it will be difficult to retrieve a particular object from among the various object in the collection when needed. This is why it is important to organise files, books and equipments, etc, in offices. Similarly the different information carriers which are normally acquired and stored in the library for use, must be organised in order to facilitate easy retrieval when the users of a library wants them.

Libraries normally acquire a large number of information carriers. In large libraries, they can run into millions of different information carriers. Thus a list showing records of all information carriers in a collection through the access point of author, title and subject and how such materials must be located must be provided. Such lists are called retrieval tools.

Taylor (1999) has identified some of the retrieval tools to include bibliographies, catalogues, indexes, finding aids and register.

*Bibliographies*: These are basically lists of information carriers delineated by subjects, form, author or geographical location. The list contains surrogates of documents such as books, journal, article, conference papers and other type of information carriers.

Each item is only listed once, usually arranged by author. Thus only one access point is provided for in a bibliography.

*Index:* The index serves a guide to the intellectual content of a document e.g. books, journal article, edited works, etc, by indicating where an item in the collection could be located. The list is generally arranged in alphabetical order by subject or name.

*Finding Aids:* These are the retrieval tools used in archives. They are called inventories, in which items in the archives are listed based on the personal or corporate sources rather than the author of the individual item.

*Registers:* They are used mainly in museums. A register contains the identity and the donor of each object in the museum. An identification number is given to each object which can be used to trace the object in a museum.

*Catalogue:* This is the list of all items in a collection, usually arranged by author, title or subject. It is an index to the collection of a library.
We shall focus mainly on this, as an example of retrieval tool since it is the major tool used for the retrieval of documents in a library collection.

## LIBRARY CATALOGUE AND CATALOGUING

The most important of the retrieval tools for organising a library collection is the catalogue. Within the context of a library, a catalogue can be defined as the list of records of the documents held in a library. However, when a catalogue records the documents (holdings) in more than one library, it is generally referred to as a Union Catalogue. The catalogue is on the other hand referred to as key or pointer to the content or holdings of a library.

### Importance of Library Catalogue

1. It enhances access to library collections.
2. It enables users to locate books on a particular subject.
3. It enables users to identify sources of information.
4. It makes it possible to identify a book with a particular author.
5. It provides full bibliographic details of a book.
6. It enables user to know whether or not certain books are in the library.
7. It shows at a glance the number of copies a library has on a title.

### Cataloguing

Cataloguing is the process of describing a book or other library item in such a way that all the important bibliographic details are highlighted. The bibliographic details include the author's name first, title of item, publisher, place of publication, date of publication,

pagination, that is, the number of pages of the book and whether there are illustrations and or diagrams, series etc.

The process of preparing entries is called cataloguing. The person who does the cataloguing is called the cataloguer.

**Types of catalogue**

- The card catalogue
- on-line (computer) public Access Catalogue I.e OPAC
- Book Catalogue
- Shelf Catalogue
- CD_ROM Catalogue
- Computer output Microform (COM) Catalogue

of all the above, the card catalogue is commonly used in most universities in the developing countries of the world like Nigeria, while the on-line public access catalogue is commonly used in the developed countries.

## CARD CATALOGUE

Each bibliographic record of an item is recorded on a 3inches by 5inches card, for every item present in the collection, there are at least three cards representing the authors, title and subject of the items. All the cards are then filled in a series of drawers' altogether constituting a catalogue cabinet. The cards can be easily interfiled. The main advantage of card catalogue is its flexibility because old and obsolete ones can be removed and new ones inserted as appropriate.

## ON-LINE PUBLIC ACCESS CATALOGUE

This is the most modern form of a catalogue. An electronic (automated) library provides its users with online access to a variety of other electronic resources (information) located both within and outside the library, it possess all the advantages of other forms of catalogue. It is flexible and can be easily updated. It is also possible to access the records through several points. Once the library is automated and connected to the outside libraries it will enable users have access to current materials like journals, books and other information resources that the user's library may not even have.

**Approaches to card catalogue and on-line public access catalogue**

The catalogue card can be approached through different headings (access points) while seeking for information in the library catalogue. For instance in an academic library, like the University library where there is divided catalogue, which indeed is divided into three, namely

- Author Catalogue Approach
- Title Catalogue Approach
- Subject Catalogue Approach

## CLASSIFICATION

Classification means a way of grouping things. Things must be well grouped and classified in order to ensure proper and unhindered accessibility to them or their effective usage. It also means placing the materials where it relates and for easy use.

According to Owolabi (1985) classification in the library context is the arrangement of publication according to subjects. It is the grouping of things or materials in a consistent logical order. It is practically impossible for anyone, librarian or documentalist to build up a systematic library without the process of classification. Classification scheme is used to classify the library materials to achieve this objective. The section is headed by the chief cataloguer.

### Types of Classification Scheme

The known types of classification scheme used in various libraries worldwide are:

- Library of congress classification scheme
- Dewey Decimal classification scheme
- Colon classification scheme
- Bibliographic classification scheme
- Moys Classification scheme
- Universal Decimal Classification Scheme

However the most commonly used classification scheme in Nigeria are the Library of congress classification scheme which is being used by academic libraries and Dewey decimal classification scheme which is being used by National and public Libraries, which are the focus of discussion in this aspect

**Library of Congress Classification Scheme (LOC)**

The library of congress classification scheme makes use of both alphabets and numerals as its symbols. The following are the basic outlines of the library of congress classification.

| | |
|---|---|
| A | General work |
| B-BJ | Philosophy, psychology |
| BL-BX | Religion |
| H-HJ | Social Sciences: Economics |
| HM-HX | Social Sciences: Sociology |
| J | Political science |
| K | Law |
| L | Education |
| P | languages and Literature |
| Q | Science |
| R | Medicine |
| S | Agriculture |
| T | Technology |
| Z | Library Science and Bibliography |

Each class is further broken down into subdivisions; the following are some of such sub-division from classes A, B, H, L, P, Q, and T

| | | |
|---|---|---|
| A | AE | Encyclopedia |
| AG | | Dictionaries |
| AN | | Newspapers |
| AS | | Academic and Learned Societies |
| AY | | Yearbooks, Almanacs and directories |
| BF | | psychology |
| H | | Social Sciences |
| J | | Political Science |
| L | | Education |
| M | | Music |
| N | | Fine Arts |
| PN | | Literature |
| PS | | American Literature |
| Q | | Science |
| R | | Medicine |

| | |
|---|---|
| S | Agriculture |
| T | Technology |
| Z | Bibliography, Library Science, Information Resources |

**Dewey decimal classification Scheme**

This scheme divides all knowledge into ten broad classes, assigning numbers to them. The ten main classes are:

| | |
|---|---|
| 000-099 | Generalities |
| 100-199 | Philosophy and related Disciplines |
| 200-299 | Religion |
| 300-399 | Social Sciences |
| 400-499 | Languages |
| 500-599 | Pure science |
| 600-699 | Technology |
| 700-799 | Arts |
| 800-899 | Literature |
| 900-999 | General Geography and History |

The main classes are further subdivided as shown below for science division:

| | |
|---|---|
| 500-509 | Science in general |
| 510-519 | Mathematics |
| 520-529 | Astronomy |
| 530-539 | Physics |
| 540-549 | Chemistry |
| 550-559 | Geology |
| 560-569 | Paleontology |
| 570-579 | Anthropology, Biology |
| 580-589 | Botany |
| 590-599 | Zoology |

**How Books Get to the Shelf in the Library**

Library materials (books and non-books) cannot get to the shelf without being duly processed. After acquisition and verification of materials, they are:

- Stamped - with the institution's ownership stamp e.g Federal University Oye-Ekiti Library
- Accessioned – given a unique serial number called accession number e.g 00045. Accession number is the unique number assigned to the library materials as soon as it

arrives the library. It is a number given to library materials serially and as such no two library materials can get the same number.

- Catalogued and classified – given the necessary bibliographic details (descriptive cataloging) I.e. author, title, author statement, edition etc
- Assigned call mark – this is a mark that directs the user to the shelf where the materials can be located. Also known as location mark or class mark. The call mark is a combination of the classification number and the cutter number e.g.:

HM
131
B49

Cutter number is an alphanumeric symbol (combination of first alphabet of the author's name or any access point, which could be a title or a corporate body, and numerals from the cutter table) added to the classification number to facilitate easier arrangement of materials in order of items within a particular class.

Other processing are:

- pasting of the typed call mark on the spine of the book
- Pasting of the date due slip
- pasting the book envelope
- insertion of the typed book card into the book envelope
- taking the necessary statistics (total number of books catalogued and classified)
- Shelving of such books appropriately on the shelves.

## SECTIONS IN THE LIBRARY

In order to achieve effective service delivery in any library most especially University Library, it is necessary to divide the library into departments and sections.

1. Administrative Department: For easy administration of a University Library, the University Librarian is appointed to head this department which is responsible to coordinate all other departments in the library. The administrative duties of the University Librarian include.

- Development of library Policies
- Coordinating the library systems
- Training of library Staff
- Maintaining Library records and financial expenditure
- Regular statistics of books catalogued
- Supervising and organising the library

2. Technical Services Department: This department is usually referred to as "behind the scene" because most their activities are performed out of sight of the library users. The section under this department includes:

- Acquisition Section: The librarian in charge of this section selects and acquire materials into the library from various sources are made and bibliographic data are scrutinized. Checking the stock of the library to avoid unnecessary duplications is ensured.
- Cataloguing Section: it is this section that information resources are catalogued and classified. Cataloguing is a process of recording materials held in the library using the bibliographic features of the materials. The catalogue is therefore, a record of materials held by the library ranging from magazine, books, journals to non-book materials.
- Serials Section: this section selects, orders, receives, organizes and shelves serial publications. These include journals, magazines, reports, newspapers, conference proceedings etc
- Reprography and Bindery Section: This section handles photocopying of materials in the library so that the needed pages of books, journal etc are photocopied for users at moderate rate.

3. Readers' Services Department: The section under this department include

I. Circulation Section: this is the section where materials in the library are charged (loaned) and discharged (returned). Shelf and shelf reading are carried out in this section also the registration of new users, sending of overdue notices and collection of fines are done in this section

II. Reserve section: Materials that are in constant and extensive demand but are short in supply in the library are kept in the reserve section. They are made available to users on request but they are not to be loaned out.

III. Reference section: Reference section provides reference and readers' advisory service to different clientele of the library. Reference materials are kept in this section for consultation but not to be borrowed.

4. Research, Documentation and Bibliographic Department: The department is normally found in the academic and research libraries. It is made up of documents, private collections, maps and manuscripts.

5. Non Print Media Resources Department: This department consists of audio visual materials like slides, transparencies, pictures, photographs and other software with their hard wares. It selects, organizes, stores and makes the information required by users available to them.

# USE OF SOFTWARE AND ELECTRONIC DATABASES IN LIBRARIES

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## 1. Introduction

The major focus of this chapter is to expose the students to the use of modern technologies in their search for information in libraries. The section discusses library software and electronic databases and how these advanced technologies can be used to **provide fast and reliable access to the resources available in the library. These software and electronic databases** makes it easy to automate library operations, thereby facilitating students in their search for information and carrying out research in their respective fields. At the end of this chapter, students should be able to:

- Define library software and electronic databases
- Identify types and examples of library software and electronic databases
- Know how these software and databases are used in their search for information and carrying out research in their respective fields.

## 2. Libraries and Electronic Databases

Traditionally, library has been defined as collection of materials, books or media that are organised for easy retrieval. The primary purpose of establishing libraries is to acquire, process, store, preserve and make available current and relevant print and electronic resources that will meet the needs of its teeming users. Access to information through Internet has transformed the role of libraries. Libraries now boast of both printed document as well as electronic information resources in their collection. The electronic documents, according to Musa, Ahmad, Yunusa and Hamisu (2015), can be stored, accessed, and delivered as and when required. Therefore, the services of libraries are not confined within the four walls of institutions, but are integrated into local, regional, national, and international networks. The use of software and electronic databases in libraries are becoming more popular due to the portability, flexibility, and instant availability of up-to-date information. Today, libraries are transforming into virtual libraries where books, journals and magazines have changed into e-books and e-journals. This has increased the global dissemination of information.

### 3. Library Software

Software is a collection of computer programs and related data that provides the instructions for telling a computer what to do and how to do it. It refers to series of instructions designed for a computer to carry out specific functions. It also refers to one or more computer programs and data held in the storage of a computer for some purposes. These are the programmes fed into the computer to control its activities. There are two main types of computer software packages: **system software** and **application software**.

- **System software** is responsible for managing variety of independent hardware components so that they can work together harmoniously. It provides the basic function for computer and helps run the computer hardware and system. It includes the combination of the following device drivers, utilities, operating system such as windows, Linux, Unix, Mac OS, Android, and servers.
- **Application software,** the second type of software, is developed to aid in any task that benefits from computation. It is the general designation of computer programs for performing tasks. It is a broad category and encompasses software of many kinds, including the Microsoft office, Corel Draw, Photoshop, internet browser, library software, business software, computer aid design, spread sheet, mathematical software, educational software, etc.

Library software therefore are computer software packages designed to perform and accomplish library functions. They have been developed to handle basic housekeeping functions of a library. Muller (2011) stated that library software are multifunction, adaptable software applications that allow libraries to manage, catalogue and circulate their materials to patrons. They enable libraries perform the primary task of cataloguing, circulation, acquisition, reference and serials management functions in an efficient manner. They are set of detailed, designed, and coded machine readable instructions that enable a computer to carry out effective library and information services or operations. The advent of software in the library makes it easy to automate library operations, thereby facilitating students and staff in higher institutions in their search for information and carrying out research in their respective fields.

Today, software have been developed for the housekeeping functions of circulation, acquisition, cataloguing, serial processing, web based OPAC system, library stock management, etc. in the library. Good and reliable library software enhances management, control and easy access to information resources. They also help to reduce time wastage in the delivery of services to the library users (Ibrahim, 2012).

### 4. Types of Library Software

There are various types of library-oriented software packages. Ukachi, Nwachukwu and Onuoha (2014) stated that library software come in two different models: **The Proprietary software** (i.e. those that require the payment of subscription fee) and the **Open Source Software** (OSS). Some of the major proprietary software products,

according to Breeding (2012), include Symphony, Millennium, Aleph, Voyager, Polaris, Library Solution, Carl X, Spydus and many others. The proprietary products have been available for many years, and have reached a high level of maturity, and remain the dominant approach used for library automation.

On the other hand, open source software is free software developed for the enhancement of library routine activities. It is not necessarily cost-free, but is free to use, free to modify, and free to share. Some of the open source software in today's market include: Emilda, EspaBiblio, Evergreen, **Greenstone, D-Space, Avanti,** Gnuteca, InfoCID, Jayuya, Koha, SLAM, SLIM, Virtua, NewGenLib. oBiblio, OPALS, OpenAmapthèque, OpenBiblio, PhpMyLibrary, PMB, Senayan etc. Koha is an example of open source software used in Federal University Oye-Ekiti.

## 5. Electronic Databases in Libraries

**An electronic database** is a collection of data, or information, specially organized for rapid search and retrieval by a computer. It is an organized collection of information. Electronic databases play significant roles as information sources in today's libraries. Uzuegbu, Chukwu and Ibegwam (2012) defined electronic databases as specialized records of related published information documents which are not available on Google or other common search engines, especially in a full text. Furthermore, Ani and Ahiauzu (2008 p.17) stated that 'electronic databases are collection of electronic information sources (e-journals or e-books) by publishers from various fields or disciplines and that some of these databases are provided free of charge to libraries in developing countries by their publishers or vendors, while others require some fees for subscription'.

Library database therefore is a searchable electronic index of published, reliable resources which provides access to a wealth of useful research materials from academic journals, newspapers, and magazines. Some databases also include e-books, relevant web resources, and various multimedia. Databases are now very popular among librarians and library users because of several factors like; speed, flexibility, wide range and the currency (BIAD, 1998). Online databases or web-based databases are widely available to library patrons in the entire world, and many patrons can tap into these databases from their own computers or electronic devices anywhere in the world. Several university libraries are subscribed to several databases online which are made available and accessible to lecturers and students of the university. Ani and Ahiauzu (2008) observed that there are available databases in some Nigerian university libraries either through free-based access or fees-based subscription. Some of these databases are subscribed with certain amount of fees while others are free and can be accessed without any subscription fees.

## 6. Examples of Library Database

**1. AGORA:** Access to Global Online Research in Agriculture is an initiative of Food

and Agriculture Organization of the United Nations (FAO) with a number of publishing partners. AGORA provides a collection of up to 6,100 key journals and 5,800 books to 2,900 institutions in more than 100 countries.

**2. EBSCOHOST:** EBSCOHOST is the leading provider of research databases, e-journals, magazine subscriptions, e-books and discovery service to libraries of all kinds.

**3. HINARI:** Health Internetwork Access to Research Initiative Programme was set up by the World Health Organization and major publishers. It provides access to one of the world's largest collections of biomedical and health literature. Up to 14,000 e-journals (in 30 different languages), 46,000 e-books, and 100 other information resources are now available to health institutions in more than 100 countries, areas and territories benefiting many thousands of health workers and researchers, and in turn, contributing to improve world health.

**4. OARE:** Online Access to Research in the Environment (OARE) is led by the United Nations Environment Programme (UNEP) in partnership with major publishers. The programme enables developing countries to gain access to one of the world's largest collections of environmental science research. Research in a wide range of disciplines is available through OARE, including Biology; Biotechnology, Genetics & Genetically Modified Species; Botany & Plant Biodiversity; Climatology, Climate Change & Meteorology; Ecology & Wildlife Conservation; Energy Conservation & Renewable Energy; Environmental Chemistry; Environmental & Natural Resource Economics; Environmental Engineering; Environmental Law, Policy & Planning; Fish & Fisheries; Forests & Forestry; Geography, Population Studies & Migration; Geology & Earth Sciences; Natural Environmental Disasters; Oceanography & Marine Biology; Pollution & Environmental Toxicology; Satellite & Remote Sensing Technologies; Soil Sciences and Desertification; Waste Management; Water, Hydrology & Wetlands; Zoology & Animal Biodiversity.

**5. DOAJ:** Direct Open Access to Journals is an online directory that indexes and provides access to high quality, open access, peer-reviewed journals. It indexes more than 10,000 open access journals covering all areas of science, technology, medicine, social science and humanities.

**6. SCIENCE DIRECT:** ScienceDirect is a database which provides subscription-based access to a large database of scientific and medical research. It hosts over 12 million pieces of content from 3,500 academic journals and 34,000 e-books.

**7. PUBMED:** Digital Archive of Biomedical and Life Sciences Journal Literature is a free full-text archive of biomedical and life sciences journal literature at the U.S. National Institutes of Health's National Library of Medicine (NIH/NLM).

**8. TEEAL:** The Essential Electronic Agriculture Library is a searchable, offline, digital library which contains mainly agriculturally focused reference journals, as well as coverage in related subject areas.

**9. PROQUEST:** ProQuest is a collection of many databases that provide access to thousands of journals, magazines, newspapers, dissertations, and other publications. ProQuest was known for its applications and information services for libraries,providing access to dissertations, theses, e-books, newspapers, periodicals, historical collections, governmental archives, cultural archives, and other aggregated databases. This content was estimated to be around 125 billion digital pages,and had commonly been accessed through library Internet gateways. This included tools for discovery and citation management, and platforms that allow library users to search, manage, use, and share research.

## 7. Conclusion

There is no doubt that the use of software and electronic databases facilitates easy access to information resources in the library. The benefits of the introduction of these resources include provision of easy access to information resources in the library, improved efficiency of library services, provision of easy access to web based online public access catalogue (OPAC), easy generation of records in the library, reduction of time in the delivery of services to the library users, among others. A concern however is that many users are not familiar with the utilization of these resources. It is apparent that the use of these electronic information resources requires special skills in Information and Communication Technologies (ICTs) that will help students navigate the information resources at their disposal. Therefore, librarians should ensure adequate use of these resources in the library by providing proper guidance and training to the users as to satisfy their needs.

## Exercises

1. Explain the term library software
2. Define two types of library software citing six (6) examples of each
3. What is an electronic database?
4. List eight (8) examples of electronic database

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# SOURCES OF INFORMATION

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## SOURCES OF INFORMATION

**Aims of the course**

At the end of this course, student should be able to know what information and information sources are, and also to have deep knowledge of different types of information sources.

**Objective of the course**

- Student should be able to define and explain what is information sources is
- Student should be able to have knowledge of different of sources information
- Should be able to distinguish the type of sources information provided by the library.
- Should be able todemonstrate the skills of getting required information from the different sources of information, primary, secondary and tertiary.

### 1. Introduction

Information is important in all human activities. Information is being generated and transmitted by individuals on daily basis. In addition, information seekers consult various sources to satisfy their needs. There are basically three sources of information: primary, secondary and tertiary sources of information. The choice of the source to consult depends on the type of information one is looking for.

### 2. WHAT IS INFORMATION

The term information is difficult to define because it has different dimensions. It means many things to different people. According to dictionary.com, information is knowledge communicated or received concerning a fact or circumstance. Oxford dictionary defined information as facts provided or learned about something or someone. Also, it can be defined as data processed, stored or transmitted by a computer.

### 3. Types of Sources of Information

- Primary sources of information

- Secondary sources of information
- Tertiary sources of information

**Primary Sources of Information**

Primary sources are original materials on which other sources of information are based. They report a discovery or share new information. First-hand accounts and information to events are presented in primary sources. Information is presented in its original form, void of various interpretation or evaluation by various authors. Usually, research outcomes and results from original research work fall into this category. Primary sources are usually the first formal appearance of results either in print or electronic format. Examples of primary sources of information are:

- Diaries
- Letters
- print speeches
- patents
- photographs
- newspapers
- Journal articles
- theses and dissertations
- survey research
- proceedings of meetings
- conferences
- original documents (birth certificates, wills, marriage licenses, trial transcripts)
- records of organizations, government agencies (treaties, annual reports etc.)
- e-mail communications
- interviews,
- video recordings,
- audio recordings,
- websites,
- artefacts, (e.g. coins, plant specimens, fossils, furniture, tools, clothing, etc.)
- works of art (e.g. painting, sculptures, musical scores, buildings, novels, poems)
- architecture,
- literature,
- music

**Secondary Sources of Information**

Secondary sources of information are created from primary sources of information. The authors (creators) of secondary sources did not have first – hand experience of the event being researched. Secondary sources give accounts of facts by describing, analyzing, interpreting, evaluating, commenting or discussing evidence provided by primary sources. It is important to note that secondary sources are not evidence, but commentary on, and discussion of evidence. For example, secondary sources of information include:

- text book
- biographical works
- commentaries
- criticisms
- dictionaries
- encyclopedia.

It is important to know and take note of how recent secondary sources of information are.

**Tertiary Sources of Information**

Tertiary sources of information act as pointers to primary and secondary sources. In fact, tertiary sources are: works that list primary and secondary resources in a specific area; as well as works which index organize and compile citations to secondary sources. Examples of tertiary sources include:

- chronologies
- bibliographies
- indexes
- encyclopedias
- directories.

All researchers have at their disposal the three basic information sources to satisfy their information needs. However, their choice of source depends on the type of information needed for a particular research

**Exercises**

1. What is information
2. List types of sources information you are familiar with
3. State and explain the differences between primary, secondary and tertiary sources of information.

# ONLINE SEARCHING: CHALLENGES, STRATEGIES AND EVALUATION

By

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## 1. Introduction

In the past few years, computers have come down in price considerably and information and communication technology are now very accessable to almost everybody. The usage of this information and communication technology to organize, process and disseminate information are still being faced with different challenges. This chapter focus on online searching, challenges encounter during online searching, the strategies for searching online and provide some basic tips that helps in evaluate online content.

## 2. Online Searching

Online searching is the process of interactively searching for and retrieving of information through the use of computer from databases that are online. This is can also mean as a search for information conducted using a computerized system; that is a specialize search using a particular string of words a characters entered into a search engine. The use of computer and communication technologies has had enormous impact on the way information is produced, organized, stored, searched and transmitted.

The search engine mechanism is a software that enables the user to query the index to return results ranked in order of relevancy. That is the programme receives users search request compares it to the entries in the index and returns the results to the user. A user must know the collection of files, though evaluation can be applied to the ranking of results. When an information seeker is using a search engine he is not searching internet live but rather the user is searching a fixed database that has been compiled by someone for the users search.

## 3. Challenges of Online Searching

The major challenges established by Klushpreet (2009) are as follows:

**Spelling Variation**: there are variations in spelling of words of the British English and American English. For example

| British | American |
|---|---|
| Honour | Honor |
| Organisation | Organization |
| Judgement | Judgment |
| Catalogue | Catalog |

**Information overload:** This is a condition in which too much information is available in online searching, particularly when the query is expressed in team that are too general. That is the mixture of relevant and non-relevant documents together in retrieval for print.

**Excessive Truncation:** This retrieves record totally different from the desired subject. If a term is truncated too much, words representing different roots will combine to form a retrieval set and it may not relate to the concept of interest. For example, searching on truncated stem 'LIB' will retrieve record containing the words: Liberty, libation, Libby, libel, Library, Librarian and Librarianship.

**Acronyms:** This is another challenge of online searching. The conflation of words to form acronyms. For example, ADD (attention deficit disorder), SAD (seasonal affective disorder) and AIDS (Acquired Immune Deficiency Syndrome) etc. All these will retrieve based on the words "add", "sad" and "aids".

**Negation:** If one is searching for documents about programmes to teach older people how to use computers, a strategy like COMPUTER AND OLDER OR ELDERLY might be a good start, but end up in retrieving documents that contain sentence like" we had hoped to include elderly people in our program but our finding wouldn't allow us to". No universal instruction for all information retrieval database.

**Homographs:** these are words that is spelled the same as another but has a different meaning. Examples include terms like cricket (a game) and cricket (an insect), KM for kilometer or knowledge management and AGF for Accountant General of the Federation or Attorney General of the Federation.

**Reverse Concept**: If one is doing a search on school management and uses the expression MANAGEMENT AND SCHOOL, one would get hints not only "school management" but also materials on "management school" which is not what was expected.

### 4. Strategies for Online Searching

There are few basic strategies that serve as a way forward that can help a user in searching online catalogues, article databases or webpages. The search strategies are:

start with giving the closet keywords to the search and mention place with it.

put the most important keywords first.

add synonyms and in free-text searching use alternative vocabulary to describe the topic.

try the same search on different search engine.

when one find a good website, check it for links to similar sites.

always be aware that there is more, just because a search term cannot be found does not mean that does not exist or available.
take advantage of advance search features.
consider subject-specific alternatives.
keep track of where one finds something useful for repeated reference.
use more than six words in a query.
truncated words to pick up singular and plural versions.
combine keywords into a phrase where possible.
distinguish concepts by parentheses.
use Boolean operators (AND OR and NOT).

### 5. Evaluation of Internet Resources

Anyone can publish anything on the internet. The internet has a lot valuable information and a lot of junk. Nobody is responsible for the credibility of the information retrieved from the internet. Here are some tips to evaluate the internet content:

title of the webpage and website
name and designation of author or creator
purpose and objectives of webpage
contact details of author
date produced and last update

### 6. Conclusion

Online Searching is widely used to find information on the internet. A variety of information sources are available on the internet and searching this vast array of information via the web is a real challenge. The searcher should select the best searching tools and search strategies to retrieve relevant information and save their time. Finally, it is worthy to note that search engines are useful mechanism for retrieving relevant information speedily from the web. However, they are not substitutes to human expertise.

### Exercise

1. What do you understand by online searching and what are the various strategies for online searching?
2. Enumerate the tips that helps in evaluate online content?
3. What are the challenges faced when searching online?

# CONSERVATION AND PRESERVATION OF LIBRARY INFORMATION RESOURCES

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1. **Introduction**
2. COURSE DESCRIPTION:

- Definition of deterioration
- Reasons for the deterioration of library resources
- Causes of deterioration
- Ways to prevent information resources from deteriorating
- Meaning of preservation and conservation
- Factors affecting preservation and conservation

**Aim of the course**

At the end, students should be able to understand deterioration, the causes of deterioration of library information resources and how to prevent them from deteriorating.

## 3. Libraries and the Need for the Conservation/Preservation of Information Resources

Library is one of the several kinds of institutions, which has the primary role of acquiring, processing, organizing as well as preserving and conserving the printed and non-printed information resources or materials for the use of present and future clienteles. In other words, the library is not only saddled with the responsibility of acquiring information resources capable of meeting the information needs of its users, it also has the duty to ensure that these resources acquired with scarce library funds are preserved and conserved such that the resources are available not only for present users of the library but also to prospective or future users. It is important to state that library information resources refer to all those media for storing information such as textbooks, journals, magazines, handbooks, directories, maps, gazette magnetic films, cassettes, optical discs, video tapes, thesis, dissertations, etc.

Deterioration of information materials is one of the basic challenges facing library materials which are prone to wear and tear, shrinkage, cracks, brittleness, warping, bio-infestation, discoloration, abrasion, hole, dust, dirt accumulation, etc. Books and other materials suffer damage or deterioration because of several groups of factors, some of which are inherent in the materials and others are beyond the control of the library. To

avoid heavy loss of materials in the library, preservation and conservation practices become extremely germane.

According to Harvey (1993), preservation refers to all managerial and financial considerations including storage and accommodation provisions, staffing levels, policies, techniques and methods involved in preserving library and archive materials and the information contained in them. Olajide (2010) also defined preservation as everything that adds to the physical wellbeing of the information resources of a library and\or archive, such that the protection, restoration and maintenance of those resources are included. Conservation on the other hand, refers to the specialized process of making fragile object safe or to a certain extent usable. VinasandVinas(1988) defined conservation as those operations which are intended to prolong the life of an object by forestalling damage and remedying deterioration.

**4. Factors that aid/cause deterioration of Library Information Resources**

Deterioration of library information resources can majorly be caused by three (3) factors or agents: **environmental factor or agent, biological factor or agent and microbiological factor or agent**. The environment where a library is situated is one of the contributing factors to the life span of information resources. If, for instance, a library is located in a swampy area, there is no doubt that reptiles and rodents would definitely find their way into the library and feast on information materials. Again, a library not far from or built in modern industrialized areas would also suffer from air pollution which would in turn damage information materials. This is because heavy concentrations of sulphur and nitrogen oxides in polluted air produce acid which hastens the process of deterioration. This is why it is very expedient to have professionals do a survey of a land which has been earmarked for the construction of a library and the environment before such library is been built.

There are two major causes of deterioration of library resources, which are **biological and microbiological factors/agents**:

- **Biological agents** include: human beings and insects (cockroaches, termites, silverfish, spider, booklice, bookworm, etc.,

These are common causes of deterioration of information materials especially the paper-based ones. Their presence in information storage areas are usually indicated by white coloring, metallic green or blue or brown, black, bright yellow and red on cover of papers. In the same vein, man is also a potential agent of information deterioration and destruction either through mutilation, theft, defacing, mishandling, bad shelving etc. In order to curtail the aforementioned deterioration, it is expedient that the library practices good hygiene within the library. Users should be educated on how to handle the

information resources of the library. Also, taking edibles to reading and\or shelve areas should be totally prohibited. Regular training on conservation/preservation and proper library hygiene should be encouraged for staff. Recalcitrant(s) should be punished so they can serve as deterrents for others.

- **Microbiological factors/agents** include fungi, bacteria, molds, etc.

These are microbial objects or bacterial that cannot be seen by human eyes except with the use of a microscope. Among the microorganism that are injurious to library materials are Bacillus sp., pseudomonas Sp., etc. Regular cleaning of books and book shelves must be encouraged in libraries to prevent dust from settling on books and their covers.

## 5. Factors Affecting Preservation and Conservation of Library Information Resources

Information managers, most especially African librarians, face the problem of effective preservation and conservation of information resources. Below are few inhibitors to effective and efficient preservation of information materials:

- **Finance**: This is very important to the provision of library services. However, there has been a downward trend in the budgetary provisions for most African libraries. This has caused lack of desired attention given to preservation and conservation of library resources by the managements of library. Many libraries do not have funds to acquire information resources capable of meeting the information needs of their users, let alone allocate certain funds to preservation and conservation of the resources.
- **Materials**: Lack of relevant and working materials and equipment contribute to the poor status of preservation and conservation of library resources.
- **Governmental economic policies**: The economic policies in most African countries do not favour libraries and other information centres. This is because most of the materials and equipment needed for conservation and preservation of information materials are not available locally. Even in cases where the equipment and materials are available locally, they are sold at exorbitant prices due to inflation rate.
- **Lack of maintenance culture**: There is usually poor maintenance of infrastructure facilities. This factor is in fact the reason for quick deterioration of collections.
- **Lack of technical know-how**: most libraries do not have staff who are equipped with the knowledge and skill of preservation and conservation. For any library to have successful preservation and conservation programmes, its staff must have relevant skills. This is because the task of preservation and conservation of information resources is a specialized field of knowledge that requires

professionals who understand the physical and chemical nature of materials in their library.

### 6. Ways of Preventing Deterioration of Library Resources

It is very important that factors that contribute or cause the deterioration of library resources are controlled/prevented to avoid partial and\or total destruction. Below are few means by which deterioration can be controlled:

- Maintenance of general hygiene and cleanliness
- Periodic inspection to storage areas
- Maintaining vigilance over new and old acquisitions
- Mechanical traps should be used in preference to toxic chemical bait. This would allow the rodents to die on the floor, wall, or ceiling.
- Air conditioner or fan should be installed to regulate the relative humidity and temperature of library storage areas at a lower degree than the favorable temperature of 25 degrees Celsius for microbial growth.
- Users should be discouraged from putting their hands in their mouth so as to get their hands wet to flip through the pages of the book. This is because carriers of contagious diseases may introduce harmful bacterial to the book.
- Appropriate and regular fumigation should be done.

### 7. Principles for Restoration of Information Resources

- The conservation and restoration manager must have skills and knowledge of conservation and restoration.
- The extent of the damage done must be known.
- Numbering of the document must be checked whether in loose or bound form.
- There must be records for all documents repaired and such records must be safely kept.
- Good inventory planning and control of the materials needed for repair and restoration of materials must be done.

In conclusion, it is no gain saying that library information resources deteriorate and there are several factors that are responsible for the quick deterioration and decay if not total destruction. On this note, it is the responsibilities of librarians to ensure that their information resources are conserved and preserved for the maximal use of their clienteles. Doing this would mean that most of the ***Ranganathan***'s Law of Library Science have been kept.

# INTRODUCTION TO INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

BY

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**OBJECTIVES;**

By the end of this lesson, students should be able to:

- Understand Computer Threats, Solutions and Preventions
- **Different types of Operating systems**
- **Understand ICT and Importance**
- **Explain Networks and Types**
- **What the Internet is**
- **State and Explain Internet of Things (IoT), Benefits, Importance and Applications in Education**
- **What is E-Commerce Blockchain, Cryptocurrencies** and NFTs

## COMPUTER THREATS

A computer threat can be "intentional" such as hacking or "accidental" such as malfunctioning of or physical damage. This can come from many ways, either human of natural disasters. Example human threat: stealing someone's account information from a trusted bank, while an example of natural disaster is when your computer is soaked in heavy rain.

### Malicious Code

Malicious code is also known as a rogue program. It is a threat to computing assets by causing undesired effects in the programmer's part. The effect is caused by an agent, with the intention to cause damage. The agent for malicious code is the writer of the code, or any person who causes its distribution. There are various kinds of malicious code. They include virus, Trojan horse, logic bomb, trapdoor and backdoor, worm and many others.

**a) Virus**

- A program that can pass on the malicious code to other programs by modifying them

- Attaches itself to the program, usually files with .doc, .xls and .exe extensions
- Destroys or co-exists with the program
- Can overtake the entire computing system and spread to other systems

**b) Trojan Horse**

- A program which can perform useful and unexpected action
- Must be installed by users or intruders before it can affect the system's assets
- An example of a Trojan horse is the login script that requests for users' login ID and password
- The information is then used for malicious Purposes.

**c) Logic Bomb**

- Logic bomb is a malicious code that goes off when a specific condition occurs.
- An example of a logic bomb is the time bomb
- It goes off and causes threats at a specified time or date

**e) Trapdoor or Backdoor**

- A feature in a program that allows someone to access the program with special privileges.
- Also known as a **backdoor**, provides a secret -- or at least undocumented -- method of gaining access to an application, operating system or online service. Programmers write trapdoors into programs for a variety of reasons. Left in place, trapdoors can facilitate a range of activities from benign troubleshooting to illegal access.

**f) Worm**

**Worm** is a standalone malware computer program that replicates itself in order to spread to other computers. It's a program that copies and spreads itself through a network. Often, it uses a computer network to spread itself, relying on security failures on the target computer to access it. Worms almost always cause at least some harm to the network, even if only by consuming bandwidth (maximum data transfer capacity), whereas viruses almost always corrupt or modify files on a targeted computer.

**Primary Differences between Worms and Viruses**

| Worms | Viruses |
|---|---|
| Operates through the network | Spreads through any medium |
| Spreads copies of itself as a Standalone program. | Spreads copies of itself as a program that attaches to other programs. |

**Security Measures:**

**1. Antivirus**

An antivirus program protects a computer against viruses by identifying and removing any computer viruses found in the computer memory, on storage media or incoming e-mail files. An antivirus program scans for programs that attempt to modify the boot program, the operating system and other programs that normally are read from but not modified. Examples are Norton Antivirus, ESET Antivirus, and Kaspersky Antivirus.

**2. Anti-Spyware**

Spyware is a program placed on a computer without the user's knowledge. It secretly collects information about the user. The spyware program communicates information to the outside source. An anti-spyware application program sometime called tracking software or a spybot is used to remove spyware. Among of the popular anti-spyware programs are:

Spybot Search and Destroy
Ad-aware
Spyware Blaster

**3. Firewall**

Firewall is a piece of hardware or software which functions in a networked environment to prevent some communications forbidden by the security policy. Firewall implements a security policy. It might permit limited access from in or outside the network perimeters or from certain user or for certain activities.

**Malicious Code Threats vs. Antivirus and Antispyware**

Security threats include virus, Trojan horse, logic bomb, trapdoor and backdoor, and worm. Antivirus and anti-spyware can be used to protect the computer from the threats by:

- limiting connectivity
- allowing only authorised media for loading data and software
- enforcing mandatory access controls
- blocking the virus from the computer program.

**Security Procedures**

**Data Protection**

We need to protect the data in the computer as it may somehow get lost or corrupted due to some viruses or mishap like fire, flood, lightning, machine failures or even human errors. There are a few ways to protect the information namely:

make backup files
detect the virus and clean the computer

warn others on virus attacks

**1) Backup Files**

**Backup**, or **data backup**, or the process of backing up, refers to the copying into an archive file of computer data that is already in secondary storage—so that it may be used to restore the original after a data loss event.

Backups are used to restore data after it has been lost due to data deletion or corruption, as well as to recover data from a previous time period depending on a user-defined data retention policy. Backups are a simple kind of disaster recovery that should be included in any disaster recovery strategy, but they should not be regarded a complete disaster recovery plan in and of itself. One reason for this is because not all backup solutions can restore data from a backup to restore a computer system or other sophisticated setup like a computer cluster, active directory server, or database server.

**2) Detect Virus and Do Cleanup**

A computer virus is able to affect and infect the way the computer works. Viruses can be detected when we run an antivirus program. We can also delete the infected files and documents.

**3) Warn Others on Virus Attack**

We can warn others on virus attacks or new viruses by sending e-mails to them.

## OPERATING SYSTEMS

1. **Microsoft Windows (Win XP, Win 2000, Win 7, 8, 8.1, 10, 10.1)**
2. **Linux**
3. **Ubuntu (10, 12, 14, 16, 17)**
4. **Fedora**
5. **Macintoch (Mac)**

## Software Applications Packages

1. **Windows (Office; Word, PowerPoint, Excel, Access…)**
2. **Ubuntu (Libre Office)**

# INTRODUCTION TO ICT

## What Is ICT?

ICT is the technology required for information processing, in particular, the use of electronic computers, communication devices and software applications to convert, store, protect, process, transmit and retrieve information from anywhere, anytime. It also refers to all the technology used to handle telecommunications, broadcast media, intelligent building management systems, audiovisual processing and transmission systems, and network-based control and monitoring functions. The term *ICT* is also used to refer to the convergence of audiovisual and telephone networks with computer networks through a single cabling or link system.

Although ICT is often considered an extended synonym for information technology (IT), its scope is broader. ICT has more recently been used to describe the convergence of several technologies and the use of common transmission lines carrying very diverse data and communication types and formats.

## NETWORK

Computer networking refers to the techniques, physical connections, and programs used to interlink two or more computers together. This is fundamentally to enable the networked computers share resources such as hard discs, modems, printers and other devices. Computer networks are the backbone of modern-day communication. Simply put, computer networking is the study of how computers can be linked to share data. Computer networking refers to connected computing devices (such as laptops, desktops, servers, smartphones, and tablets) and an ever-expanding array of **IoT** devices (such as cameras, door locks, doorbells, refrigerators, audio/visual systems, thermostats, and various sensors) that communicate with one another).

## Types of Network Connection

There are two basic types of network connections:

Physical connections and

Logical or virtual connections

**Physical connections** that let computers directly transmit and receive signals. Physical connections are defined by the medium used to carry the signal, the geometric arrangement of the computers (topology), and the method used to share information.

**Logical, or virtual,** connections that allow computer applications, such as e-mail programs and the browsers used to explore the World Wide Web, to exchange information. Logical connections are created by network protocols and allow data sharing between applications on different types of computers, such as a personal computer (PC)

running the Microsoft Corporation Windows operating system, in a network.

**Basic Components of a Network**

A classical computer network has three essential components or segments. They are as follows:

Network Software - Network software is a term that represents any software that helps a network admin perform a specific network operation with ease.

Application Software - Application software is commonly defined as any program or number of programs designed for end-users.

Network Hardware - Networking hardware includes all computers, peripherals, interface cards and other equipment needed to perform data-processing and communications within the network.

**Networking Devices**

i. **Network Interface Cards**
ii. **Hubs**
iii. **Switches**
iv. **Routers**
v. **Modems**
vi. **Uninterruptible Power Supplies**
vii. **Wiring and Cable**

Networks use three primary types of wiring (also referred to as "media"):

**Twisted-pair:** The industry standard in new installations. This wire comes in several "standards." Unshielded twisted pair (UTP)

**Coaxial:** resembles round cable TV wiring.

**Fiber-optic:** usually reserved for connections between ***"backbone"*** devices in larger networks, though in some very demanding environments, highly fault resistant fiber optic cable is used to connect desktop workstations to the network and to link adjacent buildings. Fiber-optic cable is the most reliable wiring but also the most expensive.

There are basically two types of computer networks namely:

1. Local Area Networks (LAN)
2. Wide Area Networks (WAN)

Other types of networks co-exist due to the combination of the LAN or WAN networks. Depending upon the size, computer networks are further classified Metropolitan Area Networks (MAN), Campus Area Networks (CAN) and Personal Area Networks (PAN). The topology (topology is the way the computer networks and network resources are connected) of the networking can be classified as Bus Network, Ring Network and Star Network.

**Local Area Network (LAN)** – A LAN can be defined as a collection of computers and peripherals interconnected within a limited geographical area. This area may be one building or one campus within a few kilometers. Local Area Networks speeds are much greater than Wide Area Networks.

**Wide Area Network (WAN)** - It is similar to a Local Area Network (LAN), but it's a lot bigger. Unlike LANs, WANs are not limited to a single location. Many wide area networks span long distances via telephone lines, fiber optic cables, or satellite links. They can also be composed of smaller LANs that are interconnected. The Internet could be described as the biggest WAN in the world. You could even call the Internet a Super WAN BAM if you wanted to. There are other types of WAN – (MAN, PAN and CAN).

**Campus Area Network (CAN)** - A CAN is a network that covers an educational or corporate campus. Examples include elementary schools, university campuses, and corporate buildings. A campus area network is larger than a local area network LAN since it may span multiple buildings within a specific area. Most CANs are comprised of several LANs connected via switches and routers that combine to create a single network. They operate similar to LANs; in that users with access to the network (wired or wireless) can communicate directly with other systems within the network. A college or university C.A.N. may also be called a 'residential network' or 'ResNet' or in FUOYE we call it '***fuoyeNET***' since it can only be accessed by campus residents, such as students and faculties.

**Wireless Network**

Wireless networks are computer networks that are not connected by cables of any kind. The use of a wireless network enables enterprises to avoid the costly process of introducing cables into buildings or as a connection between different equipment locations. Wireless networks use radio waves to connect devices such as laptops to the Internet, the business network and applications.

A wireless network can be added to either a LAN or a WAN to reduce cabling

Examples of wireless networking include Wi-Fi (Wireless Fidelity) and Bluetooth technology

Types of wireless communication networks

- Cellular Communication Service
- Wireless LAN (WLAN)
- Bluetooth Personal Area Network/Personal Area Network
- Wireless Data Networks

**Differences between Wired and Wireless Network**

| | Wired Network | Wireless Network |
|---|---|---|
| **Installation** | Moderate Difficulty | Easier, but beware interference |
| **Cost** | Less | More |
| **Reliability** | High | Reasonably high |
| **Security** | Reasonably good | Reasonably good |
| **Mobility** | Limited | Outstanding |
| **Performance** | Very Good | Good |

***Best Practice for a Wired Network:***

- *A wired network connection is always the most reliable and most secure internet connection.*
- *Do not use wireless printers on shared networks.*

***Best Practice for a Wireless Network:***

- *When utilizing wireless networking, utilize encrypted/secure wireless whenever possible.*

*Make sure non Wi-Fi based wireless devices operate outside the 2.4 or 5.8 GHZ ranges.*

**1. SSID**

***SSID** is short for **S**ervice **S**et **ID**entifier*. To put it another way, in layman's words, an SSID is the name for a Wi-Fi network. People typically encounter an SSID most often when they are using a mobile device to connect to a wireless network. For example, if you take your laptop to an internet café that has Wi-Fi, and attempt to connect to the local Wi-Fi network, your screen will display a list of SSIDs, these would be the names of all the networks that are within range of your mobile device. Then you can select the name of the network you would want to connect to and then enter the password (if necessary) to connect.

**For Example, the SSID for our school is: *fuoyenet***

**2. WPA and WPA2**

**WPA (Wi-Fi Protected Access)is a standard for wireless security. Your WPA2 key is an alpha-numeric security password that is required to access your internet connection. Anyone without this password cannot access your internet connection. Both WPA and WPA2 are concurrent security standards.**

## THE INTERNET

The internet is a globally connected network system that uses TCP/IP to transmit data via various types of media. The internet is a network of global exchanges – including private, public, business, academic and government networks – connected by guided, wireless and fiber-optic technologies. The terms internet and World Wide Web are often used interchangeably, but they are not exactly the same thing; the internet refers to the global communication system, including hardware and infrastructure, while the web is one of the services communicated over the internet.

The Internet is the world's largest network; the internet is made up of a community of end-users and providers. The Internet is used to Communicate, Retrieve, Research and E-Commerce. As computing advanced, peer-to-peer (P2P) communication was gradually delivered and enhanced. Since the 1990s, the internet has greatly influenced and upgraded networking to global standards. Billions of internet users rely on multiple application and networking technologies, including:

**Internet Protocol (IP):** The internet's primary component and communications backbone. Because the internet is comprised of hardware and software layers, the IP communication standard is used to address schemes and identify unique connected devices. Prominent IP versions used for communications include Internet Protocol version 4 (IPv4) and Internet Protocol version 6 (IPv6).

**Communications:** The internet is the most cost-effective communications method in the world, in which the following services are instantly available:

- Email
- Web-enabled audio/video conferencing services
- Online movies and gaming
- Data transfer/file-sharing, often through File Transfer Protocol (FTP)
- Instant messaging
- Internet forums
- Social networking
- Online shopping
- Financial services

The internet originated with the U.S. government, which began building a computer network in the 1960s known as ARPANET. In 1985, the U.S. National Science Foundation (NSF) commissioned the development of a university network backbone called NSFNET. The system was replaced by new networks operated by commercial internet service providers in 1995. The internet was brought to the public on a larger scale at around this time.

**The Web**

The web also known as the World Wide Web (WWW) provides a multimedia interface to recourses available on the internet.

Web sites are group of related pages and Web portals are designed to encourage you to visit them each time you are on the Web, to act as your home base, and to use as a gateway to their resources.

Web utilities are specialized utility programs that make using the internet and the Web easier and safer.

Browsers are programs designed to locate and display information on the World Wide Web by displaying both text and images within the page.

## INTERNET OF THINGS (IoT)

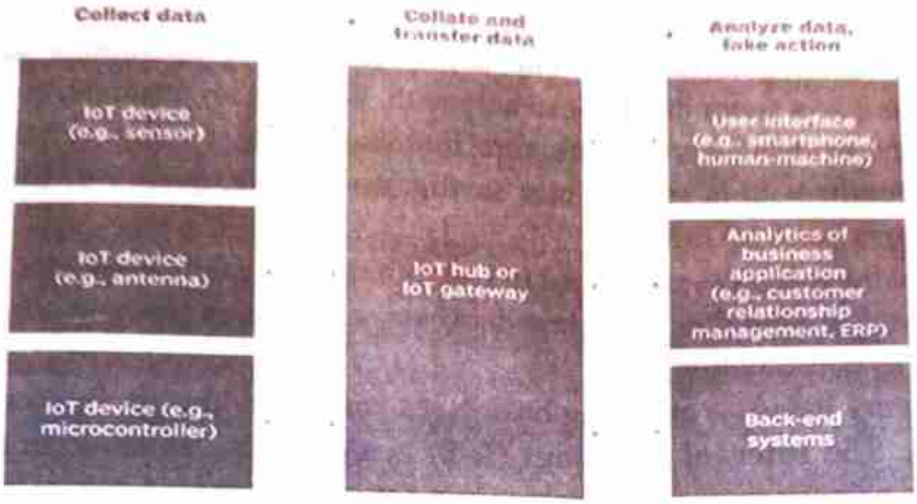
The **Internet of Things**, or IoT, is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. A *thing* in the internet of things can be a person with a heart monitor implant, a farm animal with a biochip transponder, an automobile that has built-in sensors to alert the driver when tire pressure is low or any other natural or man-made object that can be assigned an Internet Protocol (IP) address and is able to transfer data over a network.

Increasingly, organizations in a variety of industries are using **IoT** to operate more efficiently, better understand customers to deliver enhanced customer service, improve decision-making and increase the value of the business.

**How does IoT work?**

An **IoT** ecosystem consists of web-enabled smart devices that use embedded systems, such as processors, sensors and communication hardware, to collect, send and act on data they acquire from their environments. **IoT** devices share the sensor data they collect by connecting to an **IoT** gateway or other edge device where data is either sent to the cloud to be analyzed or analyzed locally. Sometimes, these devices communicate with other related devices and act on the information they get from one another. The devices do most of the work without human intervention, although people can interact with the devices -- for instance, to set them up, give them instructions or access the data.

# Example of an IoT system




**IoT System Flow**

**Why is IoT important?**

The **Internet of Things** helps people live and work smarter, as well as gain complete control over their lives. In addition to offering smart devices to automate homes, it is essential to business. **IoT** provides businesses with a real-time look into how their systems really work, delivering insights into everything from the performance of machines to supply chain and logistics operations.

**What are the benefits of IoT to organizations?**

The **Internet of Things** offers several benefits to organizations. Some benefits are industry-specific, and some are applicable across multiple industries. Some of the common benefits of **IoT** enable businesses to:

- monitor their overall business processes;
- improve the customer experience (CX);
- save time and money;
- enhance employee productivity;
- integrate and adapt business models;
- make better business decisions; and
- generate more revenue.

**IoT** encourages companies to rethink the ways they approach their businesses and gives them the tools to improve their business strategies. Generally, **IoT** is most abundant in manufacturing, transportation and utility organizations, making use of sensors and other

**IoT** devices; however, it has also found use cases for organizations within the agriculture, infrastructure and home automation industries, leading some organizations towards digital transformation.

**IoT** can benefit farmers in agriculture by making their job easier. Sensors can collect data on rainfall, humidity, temperature and soil content, as well as other factors, that would help automate farming techniques. The ability to monitor operations surrounding infrastructure is also a factor that IoT can help with. Sensors, for example, could be used to monitor events or changes within structural buildings, bridges and other infrastructure. This brings benefits with it, such as cost saving, saved time, quality-of-life workflow changes and paperless workflow.

A home automation business can utilize **IoT** to monitor and manipulate mechanical and electrical systems in a building. On a broader scale, smart cities can help citizens reduce waste and energy consumption.

**IoT** touches every industry, including businesses within healthcare, finance, retail and manufacturing. The **Internet of Things (IoT)** is a changing dynamic in the field of education. The adaptation of the digital tools is not just making education omnipresent but also it is making traditional systems of education more efficient and inclusive.

Smartphones and tablets in digital classrooms are being used by students for almost all age groups. The explanation of the complex processes via augmented reality and graphics is taking place to cater to a better understanding. Schools and educational institutes are keeping an online real-time track record of the student's progress on their portals.

Learning platforms are evolving with live classes, pre-recorded lectures and problem-solving capacities all connected to a special server and software personalized for educational use. All of this is how **IoT** has taken the education world by storm.

**IoT and Education**

The education sector is one of the most adaptive and effective in terms of deploying **IoT** devices to its usage, in order to make education more collaborative, interactive, and accessible to all. **IoT**devices give students reliable access to everything from learning materials to communication channels to good understanding, and they give teachers the ability to measure student learning progress in real-time.

Speaking of education, the COVID-19 pandemic has sufficiently highlighted the essential presence of educational sources. **IoT** just enables the shift in teaching methodology from traditional to digital with several additional benefits and increased efficiency. This can be used for teaching all the subjects ranging from languages to Mathematics to teaching practical skills like medical sciences with the use of graphics and animation to improve a better understanding of the subject matter. Not just these smart attendance devices, boards, integrated alarm systems in schools, assessment checking tools, cameras, school locks, everything can move from the physical world to

the central system-based control world and with automation.

**Applications of IoT in Education**

Technology has helped in elevating the dynamics and scope of learning, **IoT** is enabling education to become effectively practical and in certain cases surpass the boundaries of the institutes.

**1. Evolving Methodologies**

When we speak of IoT in education we primarily indicate the incorporation of digital and internet-based smart devices for the students and teachers in the educational institutes. Modern-day education platforms are adapting devices like e-books which can be downloaded and are available with zooming and saving features, smart boards instead of blackboards which can be used as a whiteboard to write with a marker and also can display topic related images and graphics to the students. Such devices are connected to a central server that can control and monitor the syllabus wise and topic-wise categorization for the students.

Not just this, voice command systems for teachers, speech to text-based note-taking systems for the students, smart security cameras, GPS tracker equipped school buses, disaster alarms and tablets, and smartphones with educational applications are changing how the traditional schools and educational systems have always operated.

These features make it safer, convenient, and accessible for the students, teachers, and parents. This is a well-understood fact that immediate transition of ways of teaching and methodologies cannot be implemented but slowly and gradually such devices are being personalized and updated with the required software. In this case, students can read traditional books and yet can benefit from IoT in the form of smart boards in the class displaying the animated and 3D versions of the topics explained to induce better learning.

**2. Automated Attendance Recording**

Attendance of the student is a concern for the teachers and in the schools that is an everyday task with no alternative. **IoT** can help in providing a solution to this hefty task of recording attendance and calculating it for various purposes. **IoT** can help in reducing this task for almost every class. Biometric attendance or barcode-based with the identity card number of the student can be used in automatically recording the attendance as they enter the classroom. In this way, there is barely any chance of discrepancy and storage.

This will not enable the teachers to devote more time to their primary concern which is teaching the students but such systems can be made more effective by sending a direct message to the pupil's parents of their absence in the classroom, making them aware of the

situation.

The same feature can be used in taking the attendance and lecture count of the teacher, the school's helping staff can also record their entry and leaving timing using their id and biometrics so that there is a clear track record of everything.

**3. Safety in Premises**

Most schools do not have the infrastructure to detect red flags for theft, abuse, sexual assault, and other crimes that can occur within the institution, nor do they have a proper contingency plan in the case of a disaster or emergency. **IoT** can help in solving such issues at a vast level, in the case of any intolerable activity that gets monitored on the camera; it can be immediately taken care of due to a network system that enables the camera recording to be displayed at various screens in the premises.

In the case of any fire or short circuit, **IoT**-based sensors can activate alarms with the exact area of the problem so that there is less hassle and danger in resolving the issue. Also, if anyone tries to break in the school smart door lock via sensors and alerts can be turned on and help can be called automatically. This will not just ensure safety but a relief in the management systems which have to face such issues once in a while.

**4. Distance Learning**

**IoT**-based systems have a feature of storing and formulating data in an application form with special software and in the form of a sign-in feature of websites that enables anyone from anywhere accessing that with a user id and a password which can be provided by the institution to their distance learning learners.

This can help each and every one who cannot be a part of a legit educational institution but still want to pursue its educational course. Live classes, pre-recorded classes, online timer-based assessment questions, tracking of time spent on the portal can all help in creating a comprehensive approach for distance learners. During the COVID19 pandemic, this has been the entire course of educational institutions, making material provide online, open booklet-based assessment has made it possible for the student to not miss out on their studies for junior classes and mid-year semesters.

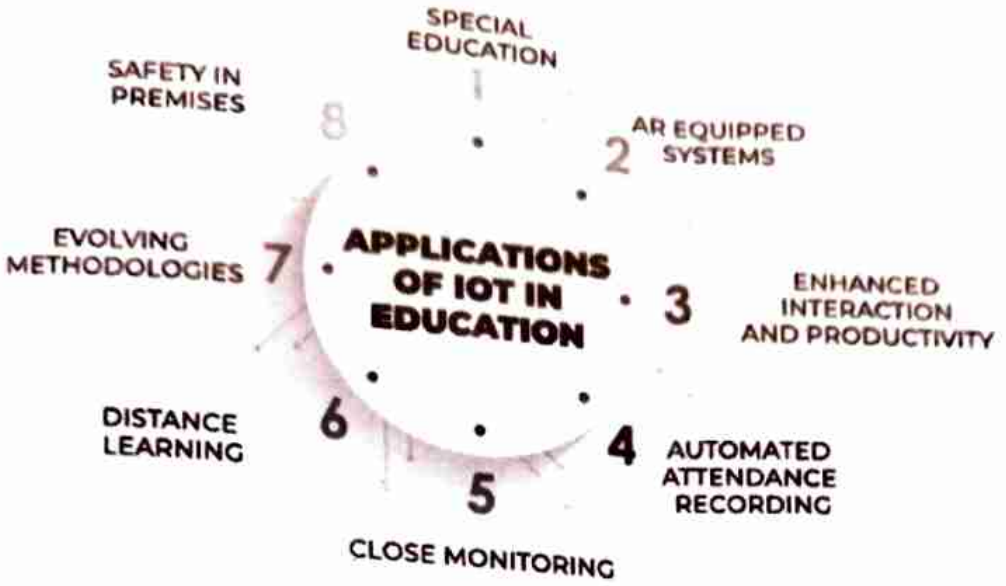



**Diagram of Application of IOT in Education**

**Some Examples of IoT in Education**

- **Scanmarker:** Via wired or wireless, Scanmarker users can quickly scan editable text from books, papers and other documents directly into a phone, tablet or computer. That text is then translatable into more than 40 languages. There's even a function that allows you to listen while you scan.
- **Blackboard:** This provides what the company describes as "a connected learning experience and support network" for a variety of fields, including K-12 and higher education.
- **Magicard:** This makes a variety of student smart cards that can be presented to an authenticating reader and linked via **IoT** to an access control system. Types of cards include ones for physical access to facilities, photo I.D. (with hologram or UV printing), time and attendance, logical access (to monitor use of electronic data like coursework, e-learning resources, printers and Internet), loyalty and membership, payment (for vending machines, printing, photocopying) and health/medical data (blood type, emergency contacts).
- **Kajeet:** Kajeet makes Wi-Fi systems for use in "smart" school buses. Increasingly, Wi-Fi enabled school buses provide students with Internet connectivity to finish and turn in homework, drivers with the ability to monitor student behavior and improve on-board safety and parent the tools to track their child's bus in real time.
- **LocoRobo:** The Loco**IoT** course instructs students on how to build, design and connect IoT systems from the ground up.

**In Conclusion, IoT solutions for education have come up with answers to enhance the quality of education across the globe by making it easy to understand and available to all. IoT is being implemented in certain schools at their own pace of adoption as it is an expensive investment.**

Pertaining to **IoT's** advantages and vast area of operation, this is worth investing in for this sector. A lot of educational platforms are recently surfacing and that's solely because **IoT**-enabled devices to come with features of extended education at convenient, easy to use, and safe platforms for the teachers and students.

**ICT Terms;**

**E-Commerce**

Buying and selling goods by the use of an internet are known as e-commerce. E-commerce helps in boosting the economy. It makes buying and selling activities easier and faster. In an e-commerce sector, customers, suppliers and employees get benefits from the usage of ICT. Customers use ICTs to connect online with suppliers to purchase products. Suppliers use ICTs to keep track of their transactions. Employees use ICTs to communicate with their customers for any inquiries.

**Blockchain technology?**

A blockchain is a decentralized ledger of all transactions across a peer-to-peer network. Using this technology, participants can confirm transactions without a need for a central clearing authority. Potential applications can include fund transfers, settling trades, voting, and many other issues.

**Cryptocurrency**

A cryptocurrency is a medium of exchange, such as the US dollar, but is digital and uses encryption techniques to control the creation of monetary units and to verify the transfer of funds.

**Bitcoin** is the name of the best-known cryptocurrency, the one for which blockchain technology was invented. To mention a few others which are; Ethereum, Tether (USDT), Litecoin, XRP, Binance Coin (BNB), Cardano, Polkadot.

**Non-Fungible Token (NFT)**

A non-fungible token (NFT) is a programmable unit of unique data that cannot be changed once it has been recorded in a distributed ledger. This is because NFTs are recorded as part of a blockchain they can be easily tracked to verify the authenticity and history of a specific digital asset.

Just like baseball cards and comic books, there's a collector's market for NFTs. In the early part of 2021, NFT marketplaces (also known as exchanges) began to attract attention from investors after Christie's auction house sold a non-fungible token for $69 million.

NFTs are created through a pay-per-use process called *minting*. Many of the online services that people use to mint NFTs also act as marketplaces where non-fungible tokens can be viewed, bought and sold.

NFT is also a well-known acronym in computing for network file transfer.

**Further Explanations on Non-Fungible Token (NFT)**

To date, companies and creators have developed a wide variety of NFTs for:

- Digital artwork
- Digital music
- Virtual real estate and other digital objects used in games
- Upselling physical consumer consumer products like sneakers
- Video clips

NFTs and Intellectual Property

It's important to remember that the person who buys an NFT is not purchasing a physical asset. Instead, they will simply own an encrypted token (symbol) in a blockchain that *represents* the original physical or digital object.

Typically, when a physical asset like a painting has been digitized and turned into an non-fungible token, the painter retains both intellectual (IP) and creative rights to the physical object, but such rights do not automatically apply to digital assets.

Because NFTs can be coded, however, developers have the ability to make sure the original artist can be paid a royalty when an NFT is sold to a new buyer. This can be an important consideration for investors and collectors.

| Original Artwork | Non-Fungible Token |
|---|---|
| Collectible exists in three dimensions and can be touched. | Collectible is composed of unique software code. |
| Financial value is based on supply and demand. | Financial value is based on supply and demand. |
| Security features include watermarks. | Security features include unique, encrypted code. |
| Is able to be altered by owner if desired. | Cannot be altered by owner for any reason. |

Major Differences between Artwork and Non-Fungible Token

How NFTs are created

As of 2021, OpenSea – an online marketplace for NFTs which use the Ethereum protocol (protocol – a basic set of rules that allow data to be be shared between computers), is one of the most popular cloud services for creating NFTs. According to their website, OpenSea is the first and largest marketplace for user-owned digital goods backed by a blockchain.

Other well-known NFT Exchanges include Mintable, Nifty Gateway and Rarible.

**Exercises:**

1. What is Computer Threats, Give two (2) examples of Computer virus and their affected areas
2. Whatare the symptoms of computer virus?
3. What are the possible solutions to computer virus?
4. Define the term 'ICT'
5. What are the major types of ICT?
6. What is the Network and Types of Network?
7. Define the Wireless Network?
8. Differences between Wired and Wireless Network?
9. What is the Internet and Internet of Things (IoT)?
10. Applications of IoT in Education
11. Examples of IoT in Education
12. Define term E-Commerce. What isBlockchain Technology and Cryptocurrency?
13. What are NFTs?

**FEDERAL UNIVERSITY OYE EKITI - Website: https://fuoye.edu.ng**

**UNIVERSITY PORTAL: This is the site where the main activities of students' registrations, courses and payment activities are done.**
**These site addresses are:**
**https://ecampus.fuoye.edu.ng**
**https://portal.fuoye.edu.ng**
**Each student is expected to visit these sites for information regularly.**

**EMAIL:**
**Your Login details: firstname.lastname.matricno@fuoye.edu.ng**
**for example, charles.ojo.12345@fuoye.edu.ng**
**Default Pwd: 'fuoye123' which will you will have to change it.**
**OR**
**Your Login details: lastname.firstname.matricno@fuoye.edu.ng**

# INFORMATION AND COMMUNICATION TECHNOLOGY (II)

by

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## 1. Introduction

Nowadays, technology has signaled a massive change on how the entire world in the 21st century view, perceives, communicate and participate effectively in the society. During the pre-colonial era, communication was achieved via the use of town crying, beating of drums, shooting of guns, burning of fires, whistling, drawing diagrams, weeping or mourning, protest or procession, etc. The advancement of technology has improved significantly the way we communicate even in the nearest possible way. In view of this, this chapter explores the introduction of information and communication technology (ICT) via the use of computer.

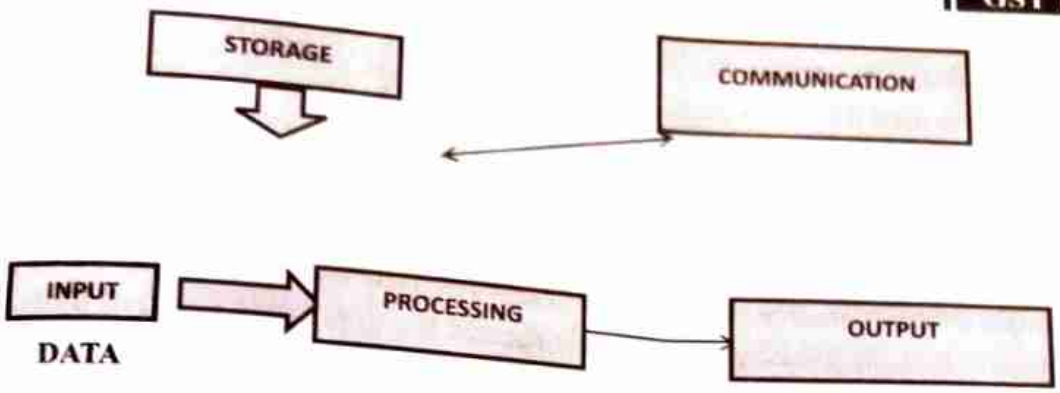
Instructional Objectives:

At the end of this chapter, students should be able to;

i. define computer
ii. discuss classification of computer
iii. discuss components of computer system
iv. explain computer virus and solutions
v. examples of fastest supercomputers

## 2. What Is Computer?

Computer is an electronic device which accepts data as input (this is done through some devices such as keyboard, mouse, joystick, camera, etc.), process (this is done through central processing unit) store (if necessary) and give out the required output (this is done through some devices such as the monitor, printer speaker, etc.) known as information (a processed data) according to specific instructions called program. It should be noted that computer gives the result of the instructions given as output. Therefore, computer users are to be careful of any data input into the computer system. Hence, the acronym GIGO which means Garbage In Garbage Out. GIGO is a concept used in mathematics and computer science meaning that the quality of the input determines the quality of the output (the data's quality aiming out cannot be better than what went in). The following are the three major operations of a computer called IPO: **Input, Process and Output**.




**The computer as an input, process, output, communication and storage (IPOCS) device**

## 1. Classification of Computer

Computer can be classified based on four criteria. These are: Generation,Types, Size and Purpose.

### 1.1. Classification of Computer according to Generation

Computer changed overtime, they improved in quality; speed, power, storage and efficiency. This means that there is a progression in the generations of discoveries characterized by a definite development. Computer generations can be classified into: pre-colonial age to 19th century and 20th century to date. A lot of attempts were made to build computing devices in the pre-computer age among which were; Abacus, slide rule, Napier's bone, Pascal's calculator, Liebniz multiplier, Jacquard's loom, Charles Babbage's analytical engine and Hollerith census machine.

**Pre-Colonial Age (19th Century)**

- **Abacus; Features and Components**

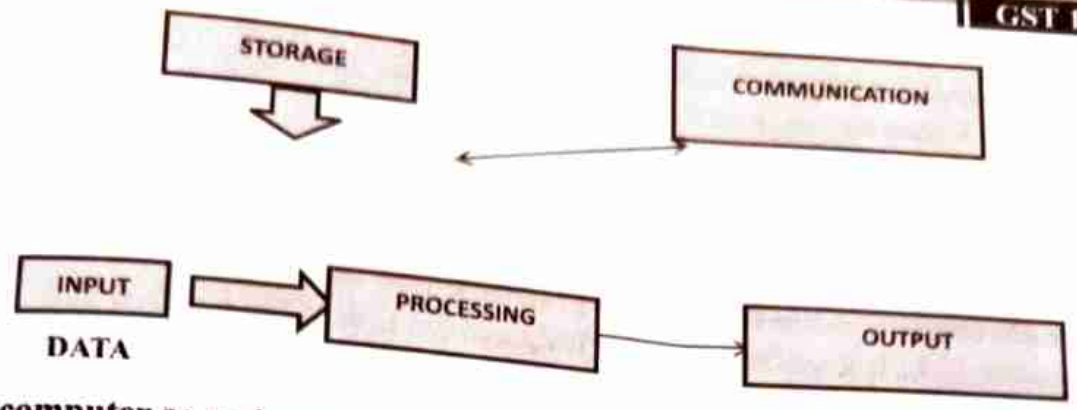
This is the first counting and calculating machine invented around 500 BC. It is made up of strings or wires, frames and beads. A modern abacus consists of rings that slide over rods, but the older one pictured around the right dates from the time when pebbles were used for counting. The modern abacus uses slides (sliding beads; the beginning of the use of 21st century slides) in columns that are divided into two by a center bar which is the real representation of five fingers in a hand. The abacus was used for arithmetic tasks such as addition and subtraction of numbers.

- **Napier's Bones: Features and Components**

Scottish mathematician and physicist John Napier (1617) noted that multiplication (asterisk) and division (forward and backward slash) of numbers could be performed by addition and subtraction respectively hence, the invention of Napier's bones.

- **Slide Rule: features and components**

Napier's invention led directly to the development of 'Slide Rule' which was first built in




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- **Slide Rule: features and components**

Napier's invention led directly to the development of 'Slide Rule' which was first built in

England in 1632. The slide rule is a mathematical analog computer used by the engineers and other mathematically professionals for generations until the invention of the pocket calculator. It is used for functions such as roots, logarithms, and trigonometry.

- **Pascal's Calculator: Features and Components**

In 1642, Blaise Pascal, at the age of 19 invented the 'Pascaline', a gear-driven calculating machine, which is used as an aid for his father who was a tax collector. Pascal built 50 gear-driven one-function calculator which could only add but really accurate. The basic mechanism of the Pascaline is a series of gears-when the first gear with ten teeth made one rotation, it shifts a second gear until it rotated ten times (one hundred times). The second shifted a third one (one thousand times) and so on. This mechanism is still in use today in the car ododmeteres, electricity meters and at the gas pumps.

- **Leibniz Multiplier: Features and Components**

This was invented eleven years after Pascal's death. Gottfried Wilhelm Leibniz was a German mathematician who was inspired by a step counting machine called pedometer and decided to build his own calculator instead of using a gear. He named his machine 'Stepped Reckonner' which could add, subtract, multiply, divide, evaluate square roots by series of additions.

- **Jacquard's Loom: Features and Components**

Jacquard's technology was a blessing to mill owners, but it put many loom operators out of work. The pattern of holes in the card determines which weaving rods could pass through. This, in turn, makes the pattern on the fabrics. Descendants of these 'Punched cards' have been in use ever since it was invented.

- **Charles Babbage Difference Engine: Features and Components**

In the early 1800s, numerical tables, such as for polynomial functions were calculated by humans. They were actually called 'computers' but was filled with human errors (syntax). In 1822, a British mathematician and inventor of named Charles Babbage invented a machine called 'Difference Engine' to calculate a series of mathematical values automatically. Babbage started to build his first engine, which was composed of around 25,000 parts, weighed 15 tons 913,600kg) and stood 8 feet (2.4m) high. It was never completed, and Babbage left to pursue another idea, a more complex 'Analytical Engine', which could be programmed using punch cards-an idea for beyond his time. His invention was considered the first mechanical computers that could calculate and print mathematical tables which were traced to be the origin of digital programmable computer. In spite of its complex design, his plan called for a basic architecture very similar to that of a modern computer hence, a father of computer.

- **Analytical Engine: Features and Components**

In 1833, Charles Babbage moved from difference engine to a general-purpose design, the analytic engine, which drew directly on Jacquard's punched cards for input and a steam engine for power, using the positions of gears and shafts to represent numbers.

The analytic engine had several features of the modern computer. The punched card was a store and a loom similar to 'memory unit' and 'central processing unit' (CPU) of modern computers. The analytical engine also had a key function that distinguishes computers from a calculator (the conditional statement). Hence, Charles Babbage was known as the father of computer!

- **Hollerith Census Machine: Features and Components**

It was invented in 1890 which was used for census successfully. It built a company (the Tabulating Machine Company) which eventually became International Business Machines, known as IBM. IBM was the biggest computer with Integrated Circuits.

**20th Century to Date**

- **ENIAC**

**ENIAC** means Electronic Numerical Integrator and Calculator. It was the first electronic general-purpose computer that has the ability to be programmed for many complex problems. It is one thousand times faster than any other machine at that time. It has the ability to multiply, divide and calculate square roots. Its components consist of vacuum tubes, ROM (Read Only Memory), parallel processing unit, patch cables and switches. It was used for addition, subtraction, multiplication, division, square root and programming.

- **EDVAC**

EDVAC simply means Electronic Discrete Variable Automatic Computer. It was a first generation computer like ENIAC which uses a single processing unit. It was the architect of modern day computers.

- **UNIVAC**

UNIVAC means Universal Automatic Computer as one of the first generation computers. It made use of metal magnetic tape and high speed tape unit for non-volatile storage. Its components include primary storage unit, magnetic tape and high-speed tape unit used for both commercial purposes, numerical and textual information processing.

- **Desktop Personal Computers**

This is a personal small micro computer system that the user can place it on the desk and work with, the name desktop personal computers. It has the system unit (otherwise known as casing contains the CPU and many other electronic components), the monitor, the keyboard and the mouse as part of its components.

- **Laptop and Notebook Computers**

These are microcomputers which are so compact that can operate by keeping it on the user's lap working on alternating current (AC) and battery power. Major features are; keyboard, LCD screen, external display, keyboard connector, batteries, built-in

pointing device, removable hard disk, etc. It is also used for various data processing and computing devices.

Notebook computer is a very small AC/battery-powered computer usually weighing less 6 lbs (2kg), which will fit inside a brief case, its features are similar to that of laptop but smaller than laptop.

- **Palmtop Computer**

These are palmtop computers that are hand-held without keyboard but the screen serves as an input device. A stylus pen is used to enter information where text can be entered manually by writing on the screen with the stylus. It can fit inside a pocket and can used for different activities.

### 1.2. Classification of Computer according to Types

Computers can be classified according to types: Analog, Digital and Hybrid.

The **analog types of computer** are mechanical, electrical, or electronic that processes data in a continuous form. They are those that measure temperatures, humidity, wind directions, voltage, volume, etc. which are characterized with scalar quantity. Examples are voltmeter, slide rule, mechanical integrators, temperature etc. Analog computers are used for scientific purpose to measure the results of experiments, or to simulate processes. It is also used for process control in industrial operations e.g., in gas stations for switching off the pump measure when a certain quantity is enriched. Analog computers are also found in engineering for structural design, space craft control, etc.

**Digital computers** are electronic devices that process data in a discrete form (ON and OFF, YES or NO). It operates on data that are expressed in binary code (using only the digits 0 and 1). A digital computer operates by taking discrete numbers, letters and symbols as separate entities and performing both mathematical and logical operations on them. It is used for commercial data processing purposes. Examples are; microcomputers such as lap tops, palm tops, desk top, etc.

**Hybrid computers** are the types of computers that have both the features of analog and digital computers. That is, the operations of analog and digital computer are performed on a given data. Examples are; mini, mainframe and super computers.

### 1.3. Classification of Computer according to Size

It should be noted that the classification of computer according to size is becoming more difficult because of the increase in the advent of technology. For now, these are the known classification: **super, mainframe, mini and microcomputers**.

- **Super computers** are designed to perform calculations as fast as current technology used to solve extremely complex problems. They are used to design automobiles, aircraft, and spacecraft in order to forecast weather and global climate. The top 10 examples are Summit (the fattest supercomputer), Sierra, Sunway TaihuLight, Tianhe-2A, PizDaint, Trinity,

AIBridgingCloudInfrastructure (ABCI), SperMUC-NG,Titan, Sequoia.

- **Mainframe computers** are high-level computers design to handle many programming languages concurrently and for the most intensive computational tasks. They are time consuming and are used heavily in both pure and applied research by scientists, the military and large businesses, for example Automated Teller Machine (ATM). They are computers with several storage devices and faster than the minicomputers. Almost all mainframe computers have the ability to run (or host) multiple operating systems. A single mainframe can replace dozens or even hundreds of smaller servers. It is a combination of memory called RAM and many processors. It also acts as a central processing unit for many workstations and terminals connected with it. They process the large and huge amount of data in petabytes.
- **Mini computers** are medium sized computers with a number of terminals (hub): (an area at which electric current enters or leaves) for input and output tasks often shared by multiple users and multi-tasking operating systems. They are very useful in super markets, airlines, cinemas and theatres.
- **Microcomputers** are relatively compact type of computer in form of desk tops, tower systems, lap tops, notebook, palm tops, etc.

### 1.4. Classification of Computer according to Purpose

**Special Purpose**: These are computers that are custom-built or tailor-made with a specific purpose only (strictly for one application). For example, a word processor is specially made for production of office documents, letters, contracts, Power points for slide presentations etc. They may include traffic control, weapon guidance systems, space invader machine, etc.

**General Purpose**: These are computers applied for a number of different applications such as businesses, scientific, educational, publishing and entertainment commonly found in homes, schools, offices, etc. They can be multi-media or internet-ready.

## 2. Components of Computer System

Computers are made up of both hardware and software components. The hardware components of the computer system are the physical parts of the computer system that can be seen, touched and felt. They are also known as the peripheral of a computer system. These are the input, storage, process and output components.

**Input Hardware**: This refers to the part of computer hardware through which data is been coded to the computer memory for processing. Examples are: keyboard, joysticks, stylus pen, mouse, etc.

**Storage Hardware**: This refers to the part of the computer hardware that keeps information for future use. The two storage units are the primary and secondary. Examples

are diskette, flash drive, pen drive, etc.

**Processing Hardware**: This refers to the part of computer hardware through which all processing task place in the central processing unit. The CPU comprises of CU (Control Unit), ALU (Arithmetic Logic Unit), and Main storage (MS).

**Output Hardware**: This refers to the unit where users get the feedback of their inputs. That is, it is a unit where the results of the operation or processing performed on data are made available to the user. The results can either be in hardcopy (from the printer) or softcopy (from the computer screen).

The software components of the computer system are the set of instructions that directs the activities of the hardware components.

**Top 10 Examples of fastest supercomputers are:**

**Number 1: Fugaku**

**Fugaku – At 442 Pflop/s, the system hit the same speed as in November. It was built by Fujitsu, has 7,630,848 cores, and resides at the RIKEN Center for Computational Science (R-CCS) in Kobe, Japan. Fugakuhas been named the fastest supercomputer.**

**Number 2: Summit**

**Summit – The fastest system in the US, Summit, scored 148.8 Pflop/s, the same as last November. It has 4,356 nodes, with two Power9 CPUs each with 22 cores each and six Nvidia Tesla V100 GPUs. The GPUs each have 80 streaming multiprocessors (SM). The nodes are linked together with a Mellanox dual-rail EDR InfiniBand network. Built by IBM, the system resides at the Oak Ridge National Laboratory (ORNL) in Tennessee.**

**Number 3: Sierra**

**Sierra – With an architecture very similar to Summit's, this system achieved 94.6 Pflop/s, the same as last time. . It is built with 4,320 nodes with two Power9 CPUs and four Nvidia Tesla V100 GPUs, and is installed at the Lawrence Livermore National Laboratory in California.**

**Number 4: Sunway TaihuLight**

**Sunway TaihuLight – This system retains its ranking with the same score it achieved with the last ranking, 93 Pflop/s. Sunway TaihuLight is powered by Sunway SW26010 processors with 10,649,600 cores. It was developed by the National Research Center of Parallel Computer Engineering & Technology (NRCPC) in China and is installed at the National Supercomputing Center in Wuxi, China.**

**Number 5: Perlmutter**

**Perlmutter – The lone new entry to the top 10 list, this system is based on the HPE Cray Shasta platform, and a heterogeneous system with both AMD EPYC-based nodes and 1536 Nvidia A100-accelerated nodes. It achieved 64.6 Pflop/s.**

**Number 6: Selene**

**Selene – Bumped down from the number five slot last time, this system an Nvidia DGX A100 SuperPOD based on an AMD EPYC processor with Nvidia A100 for acceleration and a Mellanox HDR InfiniBand as a network. It achieved 63.4 Pflop/s and is installed in-house at Nvidia facilities in the US.**

**Number 7: Tianhe-2A**

**Tianhe-2A (Milky Way-2A) – Powered by Intel Xeon CPUs and NUDT's Matrix-2000 DSP accelerators, it has 4,981,760 cores in the system to achieve 61.4 Pflop/s. It was developed by China's National University of Defense Technology (NUDT) and is deployed at the National Supercomputer Center in Guangzhou, China.**

**Number 8: JUWELS Booster Module**

**JUWELS Booster Module – At 44.1 Pflop/s, this BullSequana system is the most powerful in Europe. It's based on AMD EPYC processors with Nvidia A100s for acceleration and a Mellanox HDR InfiniBand as a network, which is similar to the Selene system. Built by Atos, it is installed at the ForschungszentrumJuelich (FZJ) in Germany.**

**Number 9: HPC5**

**HPC5 – This a PowerEdge-system build by Dell that achieves a performance of 35.5 Pflop/s using Nvidia Tesla V100s as accelerators and a Mellanox HDR InfiniBand as a network. It was installed by the Italian company EniS.p.A.**

**Number 10: Frontera**

**Frontera – This Dell C6420 achieved 23.5 Pflop/s using 448,448 Intel Platinum Xeon cores and resides at the Texas Advanced Computing Center of the University of Texas.**

**Exercises:**

1. Define the term 'Computer'
2. What are the components of computer system?
3. Explain the acronym GIGO